

Healthcare Privatization and Patient Health: Evidence from Puerto Rico ^{*}

Dayanara Diaz Vargas[†]

September 28, 2026

[Click here for recent version](#)

Abstract

Public health systems increasingly buy care from private providers: the U.S. Veterans Health Administration more than doubled such spending in 2014–2021 (Congressional Budget Office, 2021), New Zealand’s private share of publicly funded elective procedures nearly doubled from 2011–12 to 2023–24 (Radio New Zealand, 2026), and England’s health department bought £24.1 billion of care outside the NHS in 2024–25 (National Audit Office, 2026). Yet little is known about how patients fare when a publicly financed health system moves from public provision to private management. To study this shift at full scale, we examine Puerto Rico’s 1994–2000 health reform, which changed facility ownership, provider payment incentives, eligibility, and access. Exploiting the reform’s staggered rollout across counties, we find that it raised all-cause mortality by 3.7 percent. The increase is accompanied by lower ambulatory-care employment, fewer hospitals, and lower use of early prenatal care, consistent with reduced provider capacity and access to timely care as possible mechanisms. These findings suggest that a broad shift toward privately managed care may harm patient health.

JEL classification: I11, I18, H51, H75, L33.

Keywords: healthcare privatization; managed care; Puerto Rico

^{*}I thank Dr. Shantanu Khana, Dr. Mindy Marks, Dr. James Dana, Dr. Bilge Erten, Dr. Jori Barash and Dr. Elena Falcettoni for their feedback on this paper, and Urayoan G. Otero Ramirez, who gave feedback and aided in the data collection. All errors are my own.

[†]Northeastern University

1 Introduction

Health spending has risen steadily as a share of GDP across OECD countries, averaging 9.3 percent in 2024, roughly double its 1970 level, and reaching 17.2 percent in the United States (Organisation for Economic Co-operation and Development, 2025).¹ Governments finance most of this spending. Across the OECD, government and compulsory schemes cover close to three-quarters of health costs, and in the United States Medicare, Medicaid, and the Children’s Health Insurance Program (CHIP) together accounted for about 7 percent of GDP in 2024 (Centers for Medicare and Medicaid Services, 2025a; Medicaid and CHIP Payment and Access Commission, 2026). To reduce costs, governments have increasingly relied on private organizations to manage their healthcare programs. There is a rich literature that has tried to explore these effects at small scale and focusing on specific aspects of privatization. We contribute to this literature by studying this shift at an unusually broad scale examining Puerto Rico’s 1994 health reform, which transferred the management of its extensive public healthcare system to the private sector.

Before the policy took place, the Commonwealth ran a public health system of direct public provision, owning the facilities, employing salaried providers, and administering universal coverage. Between 1994 and 2000, across six staggered cohorts of the island’s 78 counties, this health reform privatized all three functions: it sold or leased its facilities to private providers, replaced provider salaries with private-sector capitated contracts, and transferred coverage to an income-based public insurance program, capped at 200 percent of the Puerto Rico poverty line and privately administered by Medicaid managed-care organizations (MCOs) (Commonwealth of Puerto Rico, 1993). In doing so, the government sought to reduce costs by shifting publicly financed care from public to private provision.

The health reform reached 78 counties in six cohorts between 1994 and 2000. Using a

¹On the CMS National Health Expenditure Accounts basis, which additionally counts the capital investment the OECD current-expenditure measure excludes, U.S. national health expenditures were 18.0 percent of GDP in 2024 (Centers for Medicare and Medicaid Services, 2025a).

county-year panel, we compare treated counties with counties that have not yet been treated, following Callaway and Sant’Anna (2021). We rely on all-cause mortality rate (measured as deaths per 100,000 county residents) as our main patient health outcome. Results indicate that mortality rate increases by 3.7 percent of its pre-treatment mean. We identify ambulatory health care services employment as a main mechanism, which indicates a decrease of 9.6 percent of its pre-treatment mean relative to the counterfactual. This is consistent with Duggan, Gupta, Jackson, and Templeton (2023), who find that hospital privatization reduces its employment levels. Results are consistent across alternative specifications and estimators.

Our main contribution is to estimate the effect of switching a public health system from public management to private. In this sense, we respond to a rising trend of governments’ reliance of private provision. For example, the U.S. government control of hospital capacity declined by 42 percent between 1983 and 2019 (Duggan et al., 2023). The Veterans Health Administration (VHA), which owns and staffs more than 1,700 sites, increased outsourced-care spending from \$7.9 billion in 2014 to \$17.6 billion in 2021, in 2021 dollars, raising its share of medical-care spending from about 12 to 20 percent (Congressional Budget Office, 2021; Panangala, Sussman, & Moreno, 2025). In England, the Department of Health and Social Care purchased £24.1 billion of healthcare outside the National Health Service in 2024–25, out of £219.2 billion in gross spending (National Audit Office, 2026). In New Zealand, private hospitals’ share of publicly funded elective procedures rose from 8.6 percent in 2011–12 to 16.7 percent in 2023–24 (Radio New Zealand, 2026). Since Puerto Rico changed ownership, payment, and access across much of its public system, we are able to provide the first causal evidence on this transfer².

Our setting also provides evidence on the transition from direct public provision to private management of publicly financed health services. The settings the literature has studied so

²Georgia’s transfer of public healthcare to private management, beginning in 2007, is studied qualitatively by Tvaliashvili, Hellowell, Roubal, Zoidze, and Clarke (2025).

far keep a public or private alternative in place (Chan, Card, & Taylor, 2023; Goodair & Reeves, 2022), privatize a set of facilities (Duggan et al., 2023; Tiemann & Schreyögg, 2012; Tvaliashvili et al., 2025), or change how providers are paid or how many patients are covered while providers remain private (Aizer, Currie, & Moretti, 2007; Duggan, 2004; Duggan, Gruber, & Vabson, 2018; Finkelstein et al., 2012). These studies leave questions on the spillover effect between private and public markets (for patients and workers). Puerto Rico instead converted its whole system at once. Additionally, with no untreated provider or beneficiary left on the island by the end of our study window, no cross-state patient or provider flows, and federal Medicaid financing that is capped (Mitchell, 2025), the design is free of cross-state spillovers and funding incentives that complicate mainland studies.

Section 2 describes the setting, Section 3 the framework, Section 4 the data, and Section 5 the empirical strategy. Section 6 presents changes in care delivery, all-cause mortality, and potential mechanisms. Section 7 examines mortality effects by cause, Section 8 assesses robustness and alternative estimation approaches, and Section 9 concludes.

2 Setting

2.1 Before the reform

From the mid-1950s, Puerto Rico managed its public healthcare program through the integrated, regionalized system described by Arbona and Ramírez de Arellano (1978). Government facilities and salaried staff provided local care through 78 Centers of Diagnostic and Treatment (CDTs), one per county. CDTs offered primary and emergency care, laboratory testing, radiology, and pharmacy services; by 1993, many also offered internal medicine, obstetrics and gynecology, and pediatrics (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). They housed vaccination, communicable-disease, and environmental-health programs, alongside federally funded maternal and child

health and agricultural migrant programs. Family Health Centers added dental care, social work, nutrition, and epidemiological nursing, with ambulatory teams comprising a physician, public-health nurse, and health assistant (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Sub-regional and regional hospitals supplied inpatient and specialized outpatient care and housed many accredited residency programs (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

Public healthcare coexisted with an extensive private healthcare sector. In 1990, the public system operated roughly 40 percent of hospital beds and served about 1.8 million residents, or 51.4% of the population (Instituto de Administración y Política de Salud, 1996). During that time, Puerto Rico had 13 public and 56 private hospitals (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). In 1992, government hospitals accounted for 39,418 of the island's 64,137 hospital births, or 61.5 percent (Puerto Rico Department of Health, 2025).

Federal Medicaid funds, available in Puerto Rico beginning in 1965 (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005), financed part of public-system expenditures, and the Commonwealth covered the remainder.³ Act 72 of 1993 attributed fiscal pressures to insufficient budgets, rising technology and supply costs, bureaucratic growth and centralization, and political interference in the Department of Health, which served residents unable to afford private care (Commonwealth of Puerto Rico, 1993). The reform sought to reduce costs for the government through contracts with insurers and providers while preserving medical and hospital access for low-income patients (Commonwealth of Puerto Rico, 1993).

³Federal Medicaid funding for Puerto Rico is subject to a capped territorial allotment (Medicaid and CHIP Payment and Access Commission, 2020).

2.2 Reform implementation

Act 72 of September 7, 1993, reorganized public healthcare around Medicaid managed care (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). It created the Puerto Rico Health Insurance Administration (ASES in Spanish) to administer coverage and negotiate insurance contracts (Commonwealth of Puerto Rico, 1993)⁴. Private MCOs managed provider networks and paid for covered services. Providers and facility operators delivered care and made staffing decisions. Public financing and government decisions over eligibility and benefits remained under this arrangement.

The government paid insurers capitated premiums, which consisted of predetermined amounts per enrollee for specified services over a fixed period. Insurers then negotiated payment terms with network providers (Commonwealth of Puerto Rico, 1993; World Health Organization & Pan American Health Organization, 1998). The goal of this system is to decrease costs, since an insurer benefits when covered care costs less than the premiums it receives. Another way in which insurers decreased costs was by setting primary-care providers as gatekeepers to specialized care, since specialist care, diagnostic tests, and procedures required their referrals (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Mental-health and pharmacy administration were subsequently contracted separately (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005) (see Figure 1, panel (d) for trends on government spending). For patients obtaining care through the program, access thus depended on self-enrollment, the insurer’s provider network, and its referral arrangements.

Alongside insurance contracting, the government transferred facilities to private operators. The 78 CDTs and eleven area and regional hospitals were sold or transferred under long-term leases; while the Commonwealth retained two of its largest hospitals, the Puerto Rico Medical Center and the Cardiovascular Center (Comisión para Evaluar el Sistema de

⁴Refer to for details on the financing of the reformed public healthcare system.

Salud del Estado Libre Asociado de Puerto Rico, 2005; Instituto de Administración y Política de Salud, 1996). Hospital births shifted toward private providers (see Figure 1, panel (b)). Government hospitals' share fell from 69 percent in 1991 to 26.6 percent in 2002, while the share of all births occurring in hospitals remained 99.6 percent.

Program eligibility required an income below 200 percent of the Puerto Rico Poverty Line, and enrollment eventually reached close to half the population (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Covered benefits included inpatient and outpatient hospital care, physician and ambulatory services, pharmacy, dental care, and mental-health and substance-abuse treatment, without pre-existing-condition exclusions. Income-dependent cost sharing ranged from \$1 to \$4 per visit and \$0.50 to \$3 per prescription (Commonwealth of Puerto Rico, 1993; Instituto de Administración y Política de Salud, 1996; World Health Organization & Pan American Health Organization, 1998).

The government selected MCOs through competitive bidding, with one MCO serving each covered area (Brugueras Fabre, 2002; World Health Organization & Pan American Health Organization, 1998). The first counties entered the insurance program in February 1994 and San Juan in 2000 (see Figure 1, panel (a), for the county rollout). The six cohorts comprise 18 counties in 1994, 27 in 1995, seven in 1996, 13 in 1997, 12 in 1998, and San Juan alone in 2000. Enrollment rose from 130,392 residents in 1994 to 1,795,067, or 47% of the population, when the rollout ended in 2000 (Administración de Seguros de Salud de Puerto Rico, 2026). Among eligible residents, take-up was 86 percent in 1995 and 1996, 93–98 percent from 1997 through 2005, and 90 percent in 2006 and 2007 (Administración de Seguros de Salud de Puerto Rico, 2026).

The 2003 coverage figures report 40 percent of residents in MCOs, 32 percent in commercial private plans, 14 percent in Medicare, and 12 percent in the public employees' plan, with 7 percent lacking a health plan (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). A separate household survey estimated 353,058 uninsured residents, or 9.10 percent of the non-institutionalized population (Comisión para Evaluar el

Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). By 2003, 118,210 of the 132,052 adults receiving mental-health treatment were treated by firms under contract with MCO's, compared with 13,842 treated by the public agency (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005). Accredited residency programs in primary-care specialties also fell from 32 to 23 over the decade, where part of that decline was due to the loss of ambulatory training sites following the privatization of hospitals and Diagnostic and Treatment Centers (Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico, 2005).

3 Theoretical framework

We adapt Hart, Shleifer, and Vishny (1997) to a bundle of healthcare services. When contracts cannot specify every production decision or dimension of care, residual control rights determine who can implement unspecified changes (S. J. Grossman & Hart, 1986; Hart et al., 1997). We represent the MCO, providers, and facility operators as one delivery manager. Control can pass through a sale, lease, or operating agreement. The model isolates this managerial decisions while holding eligibility, cost sharing, and covered benefits fixed.

For service s , the manager chooses non-contractible efforts $e_s, i_s \geq 0$. Effort e_s generates production-cost savings $c_s(e_s)$ and patient harm $b_s(e_s)$. Effort i_s generates a quality innovation with net benefit $\beta_s(i_s)$, equal to patient benefits less direct production costs. With all components measured in common value units and managerial effort costs normalized to $e_s + i_s$, the first best maximizes incremental social surplus:

$$\max_{\{e_s, i_s\}_{s \in S}} \sum_{s \in S} [c_s(e_s) - b_s(e_s) + \beta_s(i_s) - e_s - i_s]. \quad (1)$$

A private manager receiving a fixed prospective payment retains cost savings and can implement cost innovations without government approval. The manager values those savings

without bearing the associated uncontracted patient harm. Quality improvements require renegotiation; symmetric bargaining gives the manager half their net benefit. Under public management, both innovations require approval. Let $\lambda \in [0, 1]$ denote the share of innovation value requiring the incumbent manager’s participation. Because the government can obtain the remainder after replacing the manager, symmetric bargaining gives the public manager $\lambda/2$ of net innovation value. Public incentives account for patient harm but provide weaker rewards.

Assume increasing, strictly concave cost savings and net quality benefits, positive and nondecreasing marginal patient harm, $0 < \lambda \leq 1$, and interior solutions. Relative to the first best in (1),

$$e_s^G < e_s^* < e_s^P, \quad i_s^G \leq i_s^P < i_s^*, \quad (2)$$

where G and P denote public and private control, and $i_s^G = i_s^P$ when $\lambda = 1$. Private control increases cost-reducing effort and weakly increases quality-improving effort relative to public control.

We interpret our results on patient mortality rate as a potential trade off between potential harm from cost reduction and benefits from quality innovation. For example, employment levels in the healthcare sector decline if cost-reducing innovation saves more inputs than quality innovation requires. Lower employment may reflect reduced hiring, attrition, establishment exit, or layoffs. But, cost reduction efforts may reduce access or treatment intensity on labor-intensive measures that are difficult to verify. Additionally, we acknowledge that cause of death can also depend on several services and external environmental factors.

4 Data

We combine administrative and vital-statistics data for Puerto Rico’s 78 counties. The Puerto Rico Planning Board provides annual population, births, all-cause and cause-specific

deaths, and county characteristics (Junta de Planificación de Puerto Rico, 2026)⁵. Table 1 summarizes the main outcomes, and Appendix Table A1 reports baseline characteristics by cohort.

We also rely on the Quarterly Census of Employment and Wages (QCEW), which records annual employment and establishments across the U.S. (U.S. Bureau of Labor Statistics, 2025). It counts jobs covered by unemployment insurance at their workplace, excluding self-employment. We focus on Ambulatory Health Care Services (NAICS 621): physicians’ offices, outpatient clinics, and diagnostic services. The balanced sample has 59 counties with published observations throughout 1990–2000, accounting for 97.9 percent of island ambulatory employment in 1993 and 96.8 percent in 2000 (see Table 1).⁶

The Centers for Medicare and Medicaid Services Provider of Services files identify hospitals certified for Medicare, the federal insurance program for older people and certain people with disabilities (Centers for Medicare and Medicaid Services, 2025b). The 1991–2005 panel covers all 78 counties, including 47 with no certified hospitals during this period.

To extend comparisons beyond Puerto Rico’s completed rollout, we assemble mainland county mortality from CDC WONDER (National Center for Health Statistics, 2026), population and age composition from Surveillance, Epidemiology, and End Results (SEER) (National Cancer Institute, 2026), and poverty and land area from the 1990 Census (U.S. Census Bureau, 1990). The primary sample excludes states adopting Medicaid managed care by 2000.

⁵Digitized annual vital-statistics reports for 1990–2002 add births by hospital ownership, prenatal care, birth characteristics, and early observations for mortality causes incompletely recorded by the Planning Board (Puerto Rico Department of Health, 2025). We retain source flags and leave unobserved counts missing. Overlapping series have correlations of 0.99–1.00. Checks against printed subtotals and island totals assess transcription consistency; comparisons between publications assess measurement agreement, not necessarily independent registration. The data-construction appendix documents these checks, exceptions, and classification changes.

⁶Two sources assess QCEW coverage. County Business Patterns closely matches pre-reform private ambulatory employment in the available county cross-section (U.S. Census Bureau, 2000). Census microdata count more healthcare workers, with a narrowing gap over the decade (Ruggles et al., 2025). That comparison is island-wide and measures resident workers rather than workplace jobs. The data-construction appendix details differences in industry definitions, reporting, and reference periods. See Table A2 for more detail.

Mortality rates are deaths per 100,000 residents, without age adjustment, using annual intercensal population from the Planning Board in Puerto Rico and SEER on the mainland. Infant and maternal mortality use live births as denominators, per 1,000 and 100,000, respectively. Healthcare supply is per 1,000 residents; birth-process outcomes are shares of resident live births. Monetary outcomes use constant 2025 dollars and the all-items consumer price index (U.S. Bureau of Labor Statistics, 2026).

5 Identification strategy

We estimate the effects of the reform using the staggered difference-in-differences framework of Callaway and Sant’Anna (2021). For each adoption cohort, we compare changes in outcomes with those in counties not yet treated. Let Y_{it} denote county i ’s outcome in year t , G_i its adoption year, and D_{it} an indicator equal to one once county i has been treated by year t . The potential outcomes $Y_{it}(g)$ and $Y_{it}(0)$ correspond to first adopting in year g and to remaining untreated throughout. The group-time average treatment effect is $\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g]$. Write $\bar{g}$ for the last adoption year and $\mathcal{G} = \text{supp}(G) \setminus \{\bar{g}\}$. Because no county remains untreated throughout, $\mathcal{G}$ excludes the last-treated cohort, which has no valid untreated comparison group.

The ATT is then proposed by equation (2.9) Callaway and Sant’Anna (2021), imposed without covariates and without any anticipation of treatment:

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{i,g-1} \mid D_{it} = 0], \quad g \in \mathcal{G}, \quad g \leq t < \bar{g}, \quad (3)$$

where the second expectation is taken over counties not yet treated by year t , that is, those with $G_i > t$.

In our setting, the last county adopted the treatment in 2000, so $\bar{g} = 2000$. By 1998 the other 77 counties had already been treated, leaving San Juan as the only control county in

1998 and 1999. We therefore identify effects for the earlier cohorts through 1999, and write $\bar{t} = \bar{g} - 1 = 1999$ for the last identified year. San Juan is therefore only used as a control unit, since its own post-treatment effect is not identified.

For each specification we also report an event study. Let $e = t - g$ denote event time, the number of years elapsed since a county adopted the reform, and recall that G is the year in which a county is first treated. Summarizing how effects vary with length of exposure therefore amounts to averaging the $\text{ATT}(g, t)$ across cohorts at a common value of e , which is the aggregation in equation (3.4) of Callaway and Sant’Anna (2021),

$$\theta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbf{1}\{g + e \leq \bar{t}\} P(G = g \mid G + e \leq \bar{t}) \text{ATT}(g, g + e). \quad (4)$$

This is the average effect e years after adoption across the cohorts ever observed at that time period, and $e = 0$ gives the effect of the policy in the adoption year itself. The indicator restricts the sum to identified group-time effects (Callaway & Sant’Anna, 2021). The weights are cohort shares among the cohorts observed at event time e , so the set of cohorts behind $\theta_{es}(e)$ changes with e and the longest horizons rest on the 1994 and 1995 waves.

The baseline includes no covariates. Additional specifications discussed in Section 8 use the doubly robust estimand (2.7) of Callaway and Sant’Anna (2021), which allows untreated trends to depend on pre-reform county characteristics. The reported overall coefficient is θ_W^O , equation (3.10) of Callaway and Sant’Anna (2021), which uses a weighted average of all identified group-time effects with weights proportional to cohort size, so counties contributing more post-reform years receive greater total weight. Inference follows with county-specific bootstrap multipliers and 1,000 replications.

6 Results

Birth records show a shift toward private healthcare delivery, which is consistent with the policy changes discussed in Section 2.2. Relative to counties not yet treated, the government-hospital share of residents live births fell by 17.94 percentage points (s.e. 1.25), while the private-hospital share rises by 17.29 points (s.e. 1.19) (see Table 2).

We estimate an increase in all-cause mortality of 26.96 annual deaths per 100,000 residents (s.e. 10.32), equivalent to 3.7 percent of the pre-treatment mean of 724.2 (see Table 3). The parallel pre-trends assumption is satisfied as seen in its event study Figure 2 ⁷.

Event study Figure 3 shows that parallel-trends are satisfied for all pre-natal care and birth outcome estimates. The share of births receiving early prenatal care falls by 2.33 percentage points (s.e. 0.62), or 3.0 percent of its pre-reform mean of 77.4 percent (Table 4). The low-birth-weight share rises by 0.67 percentage points (s.e. 0.22), or 6.8 percent of its baseline share. The very-low-birth-weight estimate is also positive, at 6.7 percent of its baseline mean, but is not statistically distinguishable from zero.⁸ Infant and maternal mortality estimates are also statistically indistinguishable from zero. Duggan (2004) likewise finds no detectable effect of California’s Medicaid managed-care mandates on in-hospital mortality among Medicaid-insured infants. Less timely prenatal care visits to a doctor and more low-weight births are consistent with reduced access or lower treatment intensity, but could also reflect changes in eligibility or the composition of mothers giving birth.

Ambulatory healthcare supply is lower relative to the counterfactual. Employment rose from 3.58 to 6.52 per 1,000 residents between 1990 and 2000; employment in other health-care subsectors also grew.⁹ The employment effect is -0.2663 jobs per 1,000 residents (s.e. 0.1494), or -9.6 percent of the pre-reform mean of 2.773. Establishments per 1,000 decline

⁷see A3 for a visualization of the rising trend of mortality levels the years

⁸Appendix Table A4 reports annual birth outcomes for all 78 counties.

⁹Over the decade, ambulatory employment rose from 12,627 to 24,815 and establishments from 3,433 to 5,349 (Appendix Table A2, Panel C).

by 0.0490 (s.e. 0.0185), or 6.5 percent of their baseline mean of 0.7494 (Table 5). Employees per establishment increase by 0.2543 (s.e. 0.1511), or 7.6 percent of 3.356, while annual wages per worker fall by \$704 (s.e. 239), or 3.4 percent of \$20,887 in constant 2025 dollars. Certified hospitals per 1,000 residents decline by 0.0006 (s.e. 0.0002). Figure 4 reports their event studies and shows that parallel pre-trends are satisfied.

We interpret these results as indicative of a contraction in preventive healthcare services as well as specialized and emergency services. These input reductions are consistent with stronger cost reduction efforts (Section 3). Mainly, with the evidence found by (Duggan et al., 2023), where U.S. hospital privatizations also reduced personnel spending and staffing per adjusted admission. Together with higher mortality, our results are consistent with potentially harmful cost reduction efforts.

7 Heterogeneity

We examine twelve named causes of death and a residual category of unknown causes. Each outcome is annual deaths per 100,000 residents. See Table 6 and Figure 5 for estimates and their event study figures.

The cardiovascular and metabolic outcomes in Panel B show mixed effects. Diabetes mortality increases by 4.913 deaths per 100,000 (s.e. 2.409), or 9.5 percent of its pre-treatment mean, whereas hypertension mortality falls by 5.943 (s.e. 2.221), or 22.3 percent. Both conditions can benefit from regular monitoring and medication, but these opposing estimates do not support a general claim that chronic disease management improved or deteriorated. Heart-disease and cerebrovascular mortality estimates are 1.009 (s.e. 3.980) and -1.285 (s.e. 2.306), respectively; neither is statistically distinguishable from zero.

Among respiratory and infectious causes (Panel C), AIDS mortality increases by 10.22 deaths per 100,000 (s.e. 2.177), or 41.9 percent. Antiretroviral treatment is linked to lower

mortality, making treatment access a potential mechanism. Establishing that the reform disrupted this access, however, requires evidence on treatment receipt and continuity. The estimates for chronic pulmonary disease, pneumonia and influenza, and septicemia are positive but statistically indistinguishable from zero, leaving the evidence for these outcomes inconclusive.

Panel D also provides no uniform pattern across cancer and chronic liver disease. Cancer mortality falls by 9.277 deaths per 100,000 (s.e. 4.245), or 8.1 percent. The cirrhosis estimate is -0.644 (s.e. 1.245) and statistically indistinguishable from zero. Yet, attributing that reduction to earlier diagnosis or improved treatment would require evidence on disease stage, treatment receipt, and survival following diagnosis.

External causes (Panel E) raise the question of timely emergency care. Homicide mortality increases by 5.766 deaths per 100,000 (s.e. 1.974), or 41.8 percent, while the accident estimate is -1.533 (s.e. 2.433) and statistically indistinguishable from zero. Timely access to an appropriately equipped facility can matter for survival after severe injury. Additionally, the positive homicide estimate may also speak to reduced healthcare capacity. We hypothesize that emergency transport and referrals should stay within the county of incidence or neighboring ones (that may also pertain to the same cohort), but since hospitals and ambulatory healthcare services establishments decreased, commute could potentially increase and harm patients seeking immediate emergency care. However, more evidence is needed to explore this issue.

The unknown mortality causes in Panel F has the largest reported positive estimate in absolute terms with 20.60 deaths per 100,000 (s.e. 7.030), or 10.8 percent of its pre-treatment mean of 190.0. It subtracts the twelve named causes from total deaths and therefore combines the remaining causes. This statistically significant increase suggests that the reform's adverse mortality effects extended beyond the named causes. However, the category combines conditions with potentially different care needs, so the estimate cannot establish whether the increase was concentrated in a few conditions or spread across many.

More detailed cause-of-death and healthcare-use data would be needed to determine whether reduced access, treatment disruptions, or other mechanisms explain the increase.

8 Robustness

Rollout timing may be associated with county characteristics that affect mortality trends¹⁰. We therefore apply the doubly robust estimator of Callaway and Sant’Anna (2021) and condition the pre-trends across a set of relevant covariates at their baseline levels to understand the rate at which our policy affected mortality rate after accounting for any variation from these covariates. Refer to Table 7 for results and Figure 6 for event studies, and notice that all specifications satisfy the parallel pre-trends assumption. These covariates include the 1990–1993 averages of population density, crime rates, unemployment rates, and government-hospital birth counts, together with the 1990 poverty rate and population share aged 65 and over. Adjusting for each covariate separately produces statistically significant estimates between 24.69 and 27.39 additional deaths per 100,000 residents, or 3.4–3.8 percent of the pre-treatment mean, close to the baseline effect of 26.96. Including all six covariates jointly shows an increase of 35.71 additional deaths per 100,000 (s.e. 10.96), or 4.9 percent of the pre-treatment mean. The consistency of our results suggest that the observed baseline differences across counties that do not explain the estimated increase.

Table 8 examines sensitivity to the control group, population weights, and scale. To assess concerns on the differential treatment that the most populated county may have on the aggregate treatment effect, we estimate a specification excluding the capital of Puerto Rico -San Juan- from our sample. Although San Juan contributes no post-adoption effect to the baseline estimate, its mortality trajectory provides the counterfactual comparison for every effect estimated in those two years. Excluding San Juan changes the all-cause mortality

¹⁰See Table A1 for evidence of statistically significant differences between cohorts across different variables at their baseline levels. The most notable one is population density, as the rollout began at the most rural counties and ended in the most urbanized one.

estimate from 26.96 to -2.600 deaths per 100,000 residents (s.e. 12.65), an effect that is not statistically significant. One possible reason we lose precision is the loss in treated units when we exclude San Juan from the sample. San Juan is the sole not-yet-treated county in 1998 and 1999, so removing it eliminates all effects in those years. Notice that in our baseline event study (see Figure 2) mortality effects increase slowly throughout the years and becomes statistically distinguishable from zero around the specific years that we drop when we exclude San Juan. This could possibly lie in the nature of mortality rate, which is a variable that reacts with a lag to the reform. The remaining estimates cover 65 counties out of 78 in four adoption cohorts, 1994–1997; the twelve counties adopting in 1998 only serve as the counterfactual. Figure 1, Panel (C), shows the population share reached by the reform until 1997 was approximately 70%. This exclusion therefore changes both the control group, the treated cohorts and exposure years represented in the aggregate treatment effect.

Due to concerns on the differential policy effects through county population size (i.e. provider concentration, reliance in public healthcare, etc.) and relative influence of small counties (for example, small county mortality rates can be more volatile), we test an additional specification where we weight counties by their average population during their pre-treatment years (1990–1993). When we give greater influence to more populous counties, we find an increase of 32.27 additional deaths per 100,000 residents (s.e. 8.408; Table 8, column 3). Relative to the pre-treatment mean of 724.2 mortality rate, the estimate shows an increase of 4.5 percent, compared with 3.7 percent in the baseline estimation. The effect of the reform on the average county’s mortality rate remains consistent even after accounting for differences in population size.

To assess sensitivity to population normalization we estimate the reform’s effect on the absolute number of deaths. This rises as a concern that mortality rates can change because total deaths change, population changes, or both. This matters particularly if the reform itself affects population through migration. Therefore, when we estimate death counts as an effect of the reform, we find an increase of 173.2 additional deaths per county-year (s.e. 9.233),

or 54.2 percent of the pre-treatment mean of 319.3 (Table 8, column 4). A probable reason as to why total deaths increased so much as an effect of the reform is that the timing coincided with a population increase (see Table 1), while the growth of the healthcare sector contracted with lower healthcare ambulatory employment and fewer establishments per resident relative to the counterfactual (see Table 5; Figure 4). These findings suggest that the reform may have limited the healthcare system’s capacity to meet the needs of a growing population.

To complement the estimated effect in deaths per 100,000 residents, we also use a log-transformed mortality rate to express the reform’s effect approximately in percentage terms. The log-transformed specification yields 0.0247 log points (s.e. 0.0145), approximately a 2.5 percent increase in the mortality rate for counties that had the reform relative to those that have yet to adopt it (see Table 8, column 5) ¹¹.

Finally, we use mainland counties in states that had not implemented Medicaid managed care by 2000 as a comparison group. This extends the available comparisons beyond Puerto Rico’s rollout and permits estimation of effects in 2000, including San Juan’s adoption-year effect. We apply the doubly robust estimator of Callaway and Sant’Anna (2021), combining propensity-score weighting and an outcome regression to condition the control group matching by baseline population density (average population density between 1990-1993) and poverty rate (according to the 1990 U.S. Census). With 282 comparison counties, the estimate through 2000 measures 23.15 additional deaths per 100,000 residents (s.e. 20.54), or 3.2 percent of the pre-treatment mean of 731.8 (Table 8, column 6). This specification is close to the baseline estimate of 26.96 (when we only include Puerto Rico), but it is statistically indistinguishable from zero ¹².

¹¹The transformation is $\log(1 + m_{it})$, where m_{it} denotes deaths per 100,000 residents. The conversion $100[\exp(0.0247) - 1]$ equals 2.5 percent on the multiplicative scale for $1 + m_{it}$.

¹²Alternative mainland samples and estimates through 2003 are also positive and imprecise. See Appendix Table A7 and Figure ??, where we can see an increase of 21.57 deaths per 100,000 (s.e. 21.67) using 353 counties in states without Medicaid Health management Organizations (HMO’s) plans through 2000. We chose this, because HMO’s are the specific type of MCO that Puerto Rico implemented during its reform. Through 2003, the non-MMC and non-HMO estimates are 16.22 (s.e. 19.39) and 14.93 (s.e. 27.23), respectively, or 2.2 and 2.0 percent of the same baseline mean. See Appendix Table A9 for further results.

Finally, Table 9 shows a consistent mortality increase across four estimators that allow treatment effects to vary across cohorts and time. The baseline estimator of Callaway and Sant’Anna (2021) yields 26.96 additional deaths per 100,000 residents (s.e. 10.32), compared with 31.08 (s.e. 9.090) using Sun and Abraham (2021). The extended two-way fixed effects estimator of Wooldridge (2025) and the imputation estimator of Borusyak, Jaravel, and Spiess (2024) both yield 37.08, with standard errors of 7.533 and 7.320, respectively. All four estimates are statistically significant at the 1 percent level and correspond to increases of 3.7–5.1 percent of the pre-treatment mean. Our baseline estimate therefore provides the smallest point estimate among these approaches, while the alternatives support similar economic magnitudes. See Figure 8 to compare the corresponding dynamic treatment effects.

Notice that the conventional two-way fixed effects (TWFE) instead yields -1.283 deaths per 100,000 residents (s.e. 10.20), which is statistically indistinguishable from zero¹³. With staggered adoption, this specification can use already-treated counties as comparisons for counties treated later, which biases the estimation. As expressed by Borusyak et al. (2024); Callaway and Sant’Anna (2021) changes in treatment effects among those comparison counties can then contaminate the estimated effect, so the TWFE coefficient need not recover the intended average treatment effect. This limitation motivates our use of Callaway and Sant’Anna (2021) framework as the baseline.

9 Conclusion

This paper examines how Puerto Rico’s transitioned its public healthcare from public to private management. Our baseline estimates indicate that the reform increased all-cause mortality by 26.96 annual deaths per 100,000 residents, equivalent to 3.7 percent of the pre-treatment mean. These findings provide evidence of adverse population health effects during

¹³The six specifications in Appendix Table A5 show the ATT of all-cause mortality rate as estimated with TWFE holding different control variables constant. Its estimated effect ranges from -1.756 to $+3.619$, and are consistently non-statistically significant.

the transition. Findings were held consistent across different specification and identification strategies.

Two potential mechanisms explored were contractions in the healthcare labor market and reductions in health utilization, as effects of the transition. We found that employment and establishments in the Ambulatory Health Care Services industry decreased in counties that received the reform, relative to those that had not yet received it. Similarly, the share of early prenatal care visits also decreased, relative to the counterfactual. These patterns are consistent with the theoretical framework’s concern that stronger incentives to reduce costs can harm patients when relevant dimensions of healthcare are difficult to specify and enforce in contracts. The framework also allows quality improvements, and the mixed cause-specific mortality estimates may potentially indicate differences in benefits across health services.

The paper contributes evidence on the management of healthcare programs under public financing. Because the reform combined changes in facility control, payment incentives, eligibility, and access, our estimations are to be interpreted as their joint effect. Future work could also benefit from exploring the effect of fiscal savings or the reform’s net welfare effect. Longer-run comparisons would help distinguish disruption from persistent effects; variation separating policy components would help explain which features affected patients the most.

The findings nevertheless show why changes in patient health and provider capacity matter when assessing policies intended to reduce costs. For settings undertaking comparable transitions, evaluating expenditure alongside access and patient health outcomes is necessary to assess whether stronger cost-reduction incentives improve performance while preserving the benefits of publicly financed care. Therefore, before renewing managed-care contracts, we advice governments to assess whether any spending reductions are accompanied by worse patient health outcomes.

References

- Administración de Seguros de Salud de Puerto Rico. (2026). Población asegurada vs. elegible, 1994–2011 (Administrative enrollment tabulation). San Juan, PR: Administración de Seguros de Salud, Oficina de Planificación y Calidad. (Unpublished; provided to the author by ASES on 30 June 2026)
- Aizer, A., Currie, J., & Moretti, E. (2007). Does managed care hurt health? Evidence from Medicaid mothers. Review of Economics and Statistics, 89(3), 385–399. doi: 10.1162/rest.89.3.385
- Arbona, G., & Ramírez de Arellano, A. B. (1978). Regionalization of health services: The Puerto Rican experience. New York: Oxford University Press.
- Borusyak, K., Jaravel, X., & Spiess, J. (2024). Revisiting event-study designs: Robust and efficient estimation. Review of Economic Studies, 91(6), 3253–3285. doi: 10.1093/restud/rdae007
- Brugueras Fabre, A. J. (2002). Los costos de la reforma de salud de Puerto Rico (Unpublished thesis). University of Puerto Rico, Río Piedras Campus, San Juan, Puerto Rico.
- Callaway, B., & Sant’Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. Journal of Econometrics, 225(2), 200–230. doi: 10.1016/j.jeconom.2020.12.001
- Centers for Medicare and Medicaid Services. (2025a). National Health Expenditure data: Historical. Office of the Actuary, National Health Statistics Group. Retrieved from <https://www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical>
- Centers for Medicare and Medicaid Services. (2025b). Provider of Services files. Public-use files; distributed by the National Bureau of Economic Research. Retrieved from <https://www.nber.org/research/data/provider-services-files> (Last updated September 2025; accessed November 2025)
- Chan, D. C., Card, D., & Taylor, L. (2023, November). Is there a VA advantage? Evidence

- from dually eligible veterans. American Economic Review, 113(11), 3003–3043. doi: 10.1257/aer.20211638
- Comisión para Evaluar el Sistema de Salud del Estado Libre Asociado de Puerto Rico. (2005, November). Evaluación del sistema de salud de Puerto Rico: Hacia el desarrollo integral del sistema de salud de Puerto Rico (Tech. Rep.). San Juan, Puerto Rico: Estado Libre Asociado de Puerto Rico. Retrieved from <https://bvirtualogp.pr.gov/ogp/Bvirtual/reogGubernamental/PDF/Informes%20y%20Estudios/2005-11.pdf>
- Commonwealth of Puerto Rico. (1993). Ley Núm. 72 de 7 de septiembre de 1993: Ley de la Administración de Seguros de Salud de Puerto Rico [Puerto Rico Health Insurance Administration Act]. Laws of Puerto Rico, 24 L.P.R.A. § 7001 et seq. Retrieved from <https://www.asespr.org/i/beneficiarios/sobre-ases/ley-72-de-ases/>
- Congressional Budget Office. (2021, October). The Veterans Community Care Program: Background and early effects (CBO Report). Congressional Budget Office. Retrieved from <https://www.cbo.gov/publication/57583>
- Duggan, M. (2004). Does contracting out increase the efficiency of government programs? Evidence from Medicaid HMOs. Journal of Public Economics, 88(12), 2549–2572. doi: 10.1016/j.jpubeco.2003.08.003
- Duggan, M., Gruber, J., & Vabson, B. (2018). The consequences of health care privatization: Evidence from Medicare Advantage exits. American Economic Journal: Economic Policy, 10(1), 153–186. doi: 10.1257/pol.20160068
- Duggan, M., Gupta, A., Jackson, E., & Templeton, Z. S. (2023). The impact of privatization: Evidence from the hospital sector (Working Paper No. 30824). National Bureau of Economic Research. Retrieved from <https://www.nber.org/papers/w30824>
- Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (1998). Presupuesto recomendado para el año fiscal 1998–99: Departamento sombrilla de salud (presupuesto consolidado por agencia y origen de recursos). Oficina del

- Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from <https://presupuestosanteriores.pr.gov/af99/tocs/presup99.htm> (Accessed June 2026)
- Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto. (2004). Presupuesto recomendado para el año fiscal 2004–2005: Gastos operacionales por agencia (todos los orígenes de recursos). Oficina del Gobernador, Estado Libre Asociado de Puerto Rico. Retrieved from <https://presupuestosanteriores.pr.gov/af2005/index.htm> (Accessed June 2026)
- Finkelstein, A., Taubman, S., Wright, B., Bernstein, M., Gruber, J., Newhouse, J. P., ... Oregon Health Study Group (2012). The Oregon Health Insurance Experiment: Evidence from the first year. Quarterly Journal of Economics, 127(3), 1057–1106. doi: 10.1093/qje/qjs020
- Goodair, B., & Reeves, A. (2022). Outsourcing health-care services to the private sector and treatable mortality rates in England, 2013–20: An observational study of NHS privatisation. The Lancet Public Health, 7(7), e638–e646. doi: 10.1016/S2468-2667(22)00133-5
- Grossman, M. (1972). On the concept of health capital and the demand for health. Journal of Political Economy, 80(2), 223–255. doi: 10.1086/259880
- Grossman, S. J., & Hart, O. D. (1986). The costs and benefits of ownership: A theory of vertical and lateral integration. Journal of Political Economy, 94(4), 691–719. doi: 10.1086/261404
- Hart, O., Shleifer, A., & Vishny, R. W. (1997). The proper scope of government: Theory and an application to prisons. Quarterly Journal of Economics, 112(4), 1127–1161. doi: 10.1162/003355300555448
- Instituto de Administración y Política de Salud. (1996). Coordinating health care reform with the U.S. territories and possessions: The case of Puerto Rico (Final Report). San Juan, Puerto Rico: University of Puerto Rico, Medical Sciences Campus. (Prepared for the Health Care Financing Administration under grant no. 18-C-90240/2-01)

- Junta de Planificación de Puerto Rico. (2024). Informe económico al gobernador: Apéndice estadístico. Estado Libre Asociado de Puerto Rico, Oficina del Gobernador. Retrieved from <https://jp.pr.gov/apendice-estadistico-del-informe-economico-a-la-gobernador/> (Editions of 1989, 1999, 2005, 2008, 2015, 2022, and 2024. Accessed June 2026)
- Junta de Planificación de Puerto Rico. (2026). Estadísticas municipales, 1990–2010. Programa de Planificación Económica y Social, Estado Libre Asociado de Puerto Rico. Retrieved from <https://www.jp.pr.gov/planificacion-economica-social> (Workbook EstadisticasMunicipales19902010.xlsx. Accessed July 2026)
- Medicaid and CHIP Payment and Access Commission. (2020, August). Medicaid and CHIP in Puerto Rico (Fact sheet). MACPAC. Retrieved from <https://www.macpac.gov/wp-content/uploads/2020/08/Medicaid-and-CHIP-in-Puerto-Rico.pdf>
- Medicaid and CHIP Payment and Access Commission. (2026, February). MACStats: Medicaid and CHIP data book (Tech. Rep.). MACPAC. Retrieved from <https://www.macpac.gov/publication/macstats-medicaid-and-chip-data-book-2/>
- Mitchell, A. (2025, May). Medicaid financing and expenditures (CRS Report No. R42640). Washington, DC: Congressional Research Service. Retrieved from https://www.congress.gov/crs_external_products/R/PDF/R42640/R42640.13.pdf (Updated May 5, 2025. Accessed June 2026)
- National Audit Office. (2026). Department of Health and Social Care 2024-25: Overview. NAO departmental overview, January 2026. Retrieved from <https://www.nao.org.uk/overviews/department-of-health-and-social-care-2024-25/>
- National Cancer Institute. (2026). SEER U.S. county population data, 1969–2024. Surveillance, Epidemiology, and End Results Program, National Institutes of Health. Retrieved from <https://seer.cancer.gov/popdata/download.html> (Release of February 2026, Vintage 2024 estimates; file us.1990_2024.20ages.adjusted.txt.gz. Accessed June 2026)

- National Center for Health Statistics. (2026). Compressed Mortality File on CDC WONDER online database. U.S. Department of Health and Human Services, Centers for Disease Control and Prevention. Retrieved from <https://wonder.cdc.gov/cmfi-icd9.html> (ICD-10 extract, 1999–2016: <https://wonder.cdc.gov/cmfi-icd10.html>. Accessed June 2026)
- Organisation for Economic Co-operation and Development. (2025). Health at a glance 2025: OECD indicators. Paris: OECD Publishing. doi: 10.1787/8f9e3f98-en
- Pan American Health Organization. (2007, September). Health systems profile: Puerto Rico—monitoring and analyzing health systems change/reform (Tech. Rep.). Washington, D.C.: Pan American Health Organization/World Health Organization. Retrieved from <https://estadisticas.pr/files/BibliotecaVirtual/estadisticas/biblioteca/DS/DS.Perfil.Sistema.Salud-Puerto.Rico.2007.espanol.e.ingles.pdf>
- Panangala, S. V., Sussman, J. S., & Moreno, M. E. (2025, September). Department of Veterans Affairs FY2025 appropriations (CRS Report No. R48608). Congressional Research Service. Retrieved from <https://www.congress.gov/crs-external-products/R/PDF/R48608/R48608.3.pdf> (Updated September 8, 2025)
- Pillai, A., Pillai, D., Artiga, S., & Rudowitz, R. (2024, October 28). Recent changes in Medicaid financing in Puerto Rico and other U.S. territories. KFF issue brief. Retrieved from <https://www.kff.org/elections/recent-changes-in-medicaid-financing-in-puerto-rico-and-other-u-s-territories/>
- Puerto Rico Department of Health. (2025). Informe Anual de Estadísticas Vitales [annual report on vital statistics], 1990–2002. Instituto de Estadísticas de Puerto Rico (Registro Demográfico). Retrieved from <https://www.estadisticas.pr.gov/en-us/productos/informe-anual-estadisticas-vitales> (Scanned annual reports, 1990–2002. Accessed November 2025)
- Radio New Zealand. (2026, April 11). Health NZ sending more eye operations

- to private sector under long-term contracts. RNZ News. Retrieved from <https://www.rnz.co.nz/news/national/592126/health-nz-sending-more-eye-operations-to-private-sector-under-long-term-contracts> (Reports Health New Zealand administrative data on publicly funded elective procedures)
- Ruggles, S., Flood, S., Sobek, M., Backman, D., Cooper, G., Rivera Drew, J. A., ... Williams, K. C. W. (2025). IPUMS USA: Version 16.0 [dataset]. Minneapolis, MN: IPUMS. (Accessed June 2026) doi: 10.18128/D010.V16.0
- Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics, 225(2), 175–199. doi: 10.1016/j.jeconom.2020.09.006
- Tiemann, O., & Schreyögg, J. (2012). Changes in hospital efficiency after privatization. Health Care Management Science, 15(4), 310–326. doi: 10.1007/s10729-012-9193-z
- Tvaliashvili, M., Hellowell, M., Roubal, T., Zoidze, A., & Clarke, D. (2025). Private provision of health services in Georgia: A qualitative exploration of governance behaviours. BMJ Global Health, 10(12), e018922. doi: 10.1136/bmjgh-2025-018922
- U.S. Bureau of Labor Statistics. (2025). Quarterly Census of Employment and Wages. Public-use CSV data files by industry, U.S. Department of Labor. Retrieved from <https://www.bls.gov/cew/downloadable-data-files.htm> (Accessed November 2025)
- U.S. Bureau of Labor Statistics. (2026). Consumer Price Index for All Urban Consumers (CPI-U), all items, U.S. city average. Series CUUR0000SA0, U.S. Department of Labor. Retrieved from <https://data.bls.gov/timeseries/CUUR0000SA0> (Accessed June 2026)
- U.S. Census Bureau. (1990). 1990 Census of Population and Housing, Summary Tape File 3 (STF3). Decennial Census of Population and Housing, U.S. Department of Commerce. Retrieved from <https://census.gov/data/datasets/1990/dec/summary-file-3.html> (Accessed June 2026)

- U.S. Census Bureau. (2000). County Business Patterns: Puerto Rico. Annual printed series CBP/YY-53, U.S. Department of Commerce; hosted by the Puerto Rico State Data Center, Instituto de Estadísticas de Puerto Rico. Retrieved from <https://censo.estadisticas.pr/county-business-patterns> (Accessed June 2026)
- Wooldridge, J. M. (2025). Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. Empirical Economics, *69*(5), 2545–2587. doi: 10.1007/s00181-025-02807-z
- World Health Organization, & Pan American Health Organization. (1998). Reforma del sector salud: El caso de Puerto Rico (Tech. Rep.). Washington, D.C.: Pan American Health Organization. Retrieved from https://iris.paho.org/bitstream/handle/10665.2/48985/ReformaPuertoRico_spa.pdf?sequence=1

I. Tables

Table 1: Descriptive statistics by year, Puerto Rico counties

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Mortality, births and population											
All-cause mortality per 100,000	703.5 (116.7)	696.9 (103.9)	720.2 (103.9)	744.5 (108.0)	728.4 (120.9)	768.4 (132.4)	763.5 (126.5)	759.8 (114.1)	767.2 (114.3)	734.7 (109.6)	718.3 (113.2)
Government-hospital birth share (percent)	65.8 (11.3)	69.2 (33.4)	65.3 (10.9)	64.5 (11.2)	63.5 (10.4)	59.6 (11.4)	51.6 (15.2)	45.2 (15.2)	36.6 (14.6)	33.6 (18.6)	30.6 (19.9)
Private-hospital birth share (percent)	33.8 (11.3)	35.3 (18.5)	34.4 (11.0)	35.5 (10.7)	36.1 (10.6)	39.6 (11.7)	48.1 (15.3)	54.6 (15.3)	63.1 (14.6)	66.2 (18.6)	69.2 (19.9)
Population	45,228 (59,205)	45,502 (59,564)	45,884 (60,064)	46,430 (60,778)	47,253 (61,281)	47,684 (59,393)	47,240 (58,898)	47,638 (58,908)	48,053 (58,930)	48,489 (58,966)	48,828 (59,001)
Unemployment rate (percent)	17.56 (4.56)	17.78 (4.99)	18.99 (4.95)	19.68 (5.11)	18.83 (4.94)	16.08 (4.40)	16.23 (4.57)	15.57 (4.60)	16.16 (4.70)	14.90 (4.55)	13.24 (4.15)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Panel B: Ambulatory Health Care Services											
Employment per 1,000 residents	2.24 (2.15)	2.60 (2.96)	2.75 (2.79)	2.90 (2.70)	3.20 (3.02)	3.27 (2.78)	3.71 (2.96)	4.18 (3.32)	4.47 (3.59)	4.68 (3.47)	4.83 (3.56)
Establishments per 1,000 residents	0.670 (0.473)	0.705 (0.479)	0.765 (0.515)	0.773 (0.513)	0.782 (0.518)	0.828 (0.564)	0.887 (0.576)	0.944 (0.601)	0.990 (0.616)	1.065 (0.640)	1.089 (0.657)
Employees per establishment	3.11 (1.37)	3.28 (1.70)	3.27 (1.60)	3.46 (1.54)	3.81 (1.67)	3.77 (1.40)	4.01 (1.46)	4.20 (1.50)	4.29 (1.69)	4.22 (1.47)	4.27 (1.44)
Wages per worker (2025 USD)	20,104 (5,470)	20,134 (5,170)	20,821 (5,251)	21,283 (5,107)	22,031 (4,945)	22,796 (4,843)	22,762 (4,238)	23,512 (4,324)	23,702 (4,678)	23,848 (4,145)	23,742 (4,729)
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
% of employment observed	97.9	97.8	97.8	97.9	97.7	97.0	96.9	96.8	96.4	96.5	96.8
Panel C: Medicare-certified providers											
Hospitals per 1,000 residents		0.018 (0.026)	0.018 (0.027)	0.018 (0.027)	0.018 (0.027)	0.018 (0.026)	0.018 (0.026)	0.017 (0.026)	0.017 (0.026)	0.018 (0.026)	0.017 (0.025)
Hospital beds per 1,000 residents		1.55 (2.55)	1.57 (2.58)	1.56 (2.57)	1.54 (2.54)	1.54 (2.56)	1.55 (2.56)	1.53 (2.54)	1.52 (2.53)	1.60 (2.65)	1.52 (2.48)
Treated counties		0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties		78	78	78	60	33	26	13	1	1	0
% of population observed		100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Sources: Panel A, the Puerto Rico Planning Board’s municipal series; Panel B, the Bureau of Labor Statistics Quarterly Census of Employment and Wages; Panel C, the Medicare Provider of Services file. Employees per establishment is county employment over county establishments; wages are annual payroll over average annual employment, in 2025 dollars; the government-hospital birth share is births in a government-operated hospital divided by all births.

Table 2: Effect of the Health Reform on Utilization Between Types of Providers

	(1) Government-hospital births	(2) Private-hospital births
ATT	-0.1794*** (0.0125)	0.1729*** (0.0119)
Pre-treatment mean	0.6418	0.3642
ATT as % of pre-treatment mean	-28.0%	47.5%
Treated counties	78	78

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Outcomes are share of births. Dataset is balanced.

Table 3: Effect of the Health Reform on All-Cause Mortality

	(1) All causes per 100,000
ATT	26.96*** (10.32)
Pre-treatment mean	724.2
ATT as % of pre-treatment mean	3.7%
Treated counties	78

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Dataset is unbalanced.

Table 4: Effect of the Health Reform on Prenatal Care and Prenatal-Care-Sensitive Outcomes

	(1)	(2)	(3)	(4)	(5)
	Infant mortality per 1,000 births	Maternal mortality per 100,000 births	Low birth weight share of births	Very low birth weight share of births	Early prenatal care share of births
ATT	0.0985 (0.7941)	-4.291 (22.43)	0.0067*** (0.0022)	0.0008 (0.0010)	-0.0233*** (0.0062)
Pre-treatment mean	12.56	19.27	0.0987	0.0121	0.7735
ATT as % of pre-treatment mean	0.8%	-22.3%	6.8%	6.7%	-3.0%
Treated counties	78	78	78	78	78

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Births very low bw is unavailable for 1992; births low bw is unavailable for 1992; births early prenatal is unavailable for 1992. Each covers all 78 counties in the years it appears. Dataset is unbalanced.

Table 5: Effect of the Health Reform on Health Supply

	(1) Employment per 1,000 residents	(2) Establishments per 1,000 residents	(3) Employees per establishment ratio	(4) Wages per worker 2025 USD	(5) Certified hospitals per 1,000 residents
ATT	-0.2663* (0.1494)	-0.0490*** (0.0185)	0.2543* (0.1511)	-703.9*** (238.9)	-0.0006*** (0.0002)
Pre-treatment mean	2.773	0.7494	3.356	20,886.5	0.0166
ATT as % of pre-treatment mean	-9.6%	-6.5%	7.6%	-3.4%	-3.4%
Treated counties	59	59	59	59	78

Notes: Callaway and Sant'Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Dataset is balanced.

Table 6: Effect of the Health Reform on Mortality by Cause

<i>Panel A. All-cause mortality</i>				
	(1)			
	All causes			
ATT	26.96*** (10.32)			
Pre-treatment mean	724.2			
ATT as % of pre-treatment mean	3.7%			
Treated counties	78			
<i>Panel B. Cardiovascular and metabolic causes</i>				
	(2)	(3)	(4)	(5)
	Heart disease	Cerebrovascular	Hypertension	Diabetes
ATT	1.009 (3.980)	-1.285 (2.306)	-5.943*** (2.221)	4.913** (2.409)
Pre-treatment mean	156.6	36.72	26.68	51.55
ATT as % of pre-treatment mean	0.6%	-3.5%	-22.3%	9.5%
Treated counties	78	78	78	78
<i>Panel C. Respiratory and infectious causes</i>				
	(6)	(7)	(8)	(9)
	Chronic pulmonary	Pneumonia and influenza	Septicemia	AIDS
ATT	1.766 (2.205)	0.9216 (2.363)	2.097 (1.362)	10.22*** (2.177)
Pre-treatment mean	29.12	29.44	11.31	24.39
ATT as % of pre-treatment mean	6.1%	3.1%	18.5%	41.9%
Treated counties	78	78	78	78
<i>Panel D. Cancer and chronic liver disease</i>				
	(10)	(11)		
	Cancer	Cirrhosis		
ATT	-9.277** (4.245)	-0.6440 (1.245)		
Pre-treatment mean	114.4	18.64		
ATT as % of pre-treatment mean	-8.1%	-3.5%		
Treated counties	78	78		
<i>Panel E. External causes</i>				
	(12)	(13)		
	Accidents	Homicide		
ATT	-1.533 (2.433)	5.766*** (1.974)		
Pre-treatment mean	36.02	13.78		
ATT as % of pre-treatment mean	-4.3%	41.8%		
Treated counties	78	78		
<i>Panel F. Unclassified</i>				
	(14)			
	Unclassified			
ATT	20.60*** (7.030)			
Pre-treatment mean	190.0			
ATT as % of pre-treatment mean	10.8%			
Treated counties	78			

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Deaths pulmonary is unavailable for 1995; deaths homicide is unavailable for 1995; deaths septicemia is unavailable for 1990, 1991, 1992, 1993, 1995; deaths aids is unavailable for 1990, 1991, 1995. Each covers all 78 counties in the years it appears. Unclassified is all-cause deaths less the causes printed above it, so it closes the decomposition rather than naming a further cause. Each cause below is the following share of the pre-treatment all-cause mean: diabetes 7%, heart disease 22%, cerebrovascular 5%, cancer 16%, hypertension 4%, cirrhosis 3%, pneumonia and influenza 4%, septicemia 2%, chronic pulmonary 4%, aids 3%, homicide 2%, accidents 5%, unclassified 26%. Outcomes are per 100,000. Dataset is unbalanced.

Table 7: Alternative Specifications (with Conditional Trends)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>All-cause mortality</i>								
ATT (per 100,000)	26.96*** (10.32)	25.52** (10.69)	26.84** (10.44)	26.52** (10.53)	24.69** (10.19)	26.01** (10.88)	27.39*** (10.41)	35.71*** (10.96)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	3.7%	3.5%	3.7%	3.7%	3.4%	3.6%	3.8%	4.9%
<i>Covariates</i>								
Population density		✓						✓
Age 65 and over share			✓					✓
Crime				✓				✓
Poverty rate					✓			✓
Unemployment rate						✓		✓
Pre-reform government births							✓	✓
Treated counties	78	78	78	78	78	78	78	78
Estimated through	1999	1999	1999	1999	1999	1999	1999	1999
Effective comparison counties	.	20.28	25.28	21.35	8.659	12.17	19.49	3.173
Group-time cells dropped	0	0	0	0	0	0	0	0

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Each control variable is one number per county. 1990 to 1993 means: population density, crime, unemployment rate and pre-reform government births. 1990 Census values: age 65 and over share and poverty rate. Density is residents per square mile; the age 65 and over share is the 1990 Census counts aged 65 to 84 and 85 and over, over total county population; crime is total reported offences per 1,000 residents, murders included; the poverty rate is the fraction of families below the Census poverty threshold, of all families; unemployment is the county annual rate; pre-reform government births is a count of births in government-operated hospitals and not a share of births. Dataset is balanced. Conditional columns reweight the comparison group by the estimated odds of cohort membership at each cohort’s base period; 72 counties fall outside the 0.05 to 0.95 range of that probability in column (8), where the weights are least even.

Table 8: Alternative Specifications

	(1) Baseline	(2) Excluding San Juan	(3) Population weighted	(4) Levels	(5) Logs	(6) Never-treated
<i>All-cause mortality</i>						
ATT	26.96*** (10.32)	-2.600 (12.65)	32.27*** (8.408)	173.2*** (9.233)	0.0247* (0.0145)	23.15 (20.54)
Pre-treatment mean	724.2	724.2	724.2	319.3	6.576	731.8
ATT as % of pre-treatment mean	3.7%	-0.4%	4.5%	54.2%	2.5%	3.2%
Treated counties	78	77	78	78	78	78
Never-treated counties						282

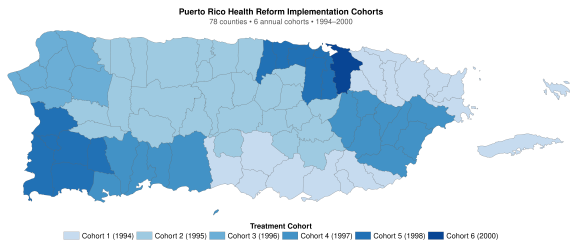
Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. The outcome is measured per 100,000. Every column is the Callaway and Sant’Anna estimator. The never-treated column is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), the same covariates as the never-treated tables. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Dataset is balanced.

Table 9: Alternative Estimators

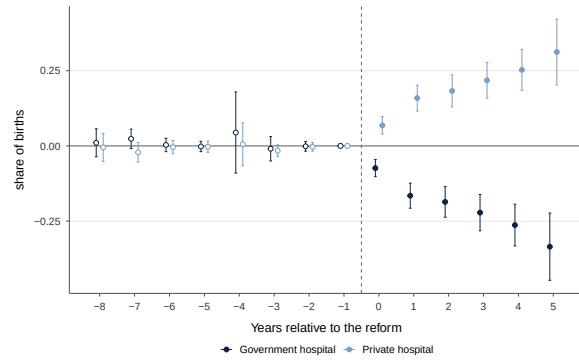
	(1) Callaway & Sant’Anna	(2) TWFE	(3) Sun & Abraham	(4) ETWFE	(5) Borusyak, Jaravel & Spiess
<i>All-cause mortality</i>					
ATT	26.96*** (10.32)	-1.283 (10.20)	31.08*** (9.090)	37.08*** (7.533)	37.08*** (7.320)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2
ATT as % of pre-treatment mean	3.7%	-0.2%	4.3%	5.1%	5.1%
Treated counties	78	78	78	78	78

Notes: TWFE is two-way fixed effects OLS and ETWFE is the extended two-way fixed effects estimator of Wooldridge (2025). The named columns are Callaway and Sant’Anna (2021), Sun and Abraham (2021), and the imputation estimator of Borusyak, Jaravel and Spiess (2024). Callaway and Sant’Anna is the baseline identification strategy. Standard errors in parentheses: for CS, county-clustered multiplier bootstrap, 1,000 replications; for TWFE, SA, ETWFE and BJS, clustered on the county. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. The outcome is measured per 100,000. Dataset is unbalanced.

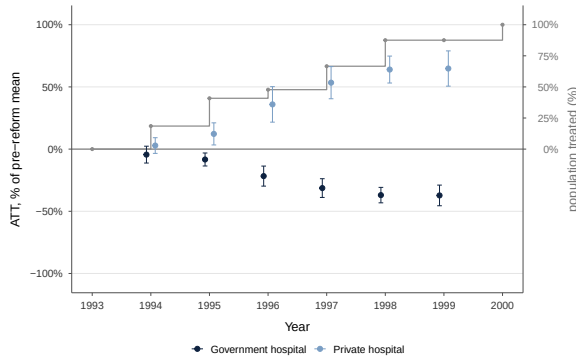
II. Figures



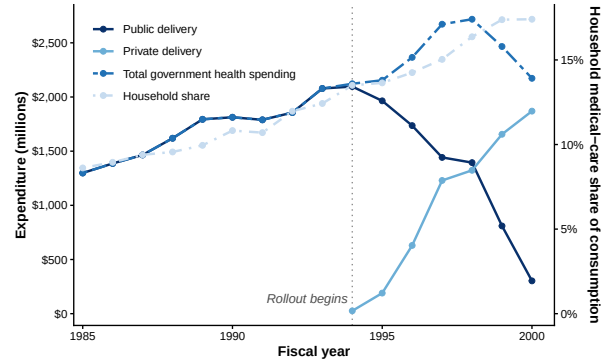
(a) Adoption cohorts



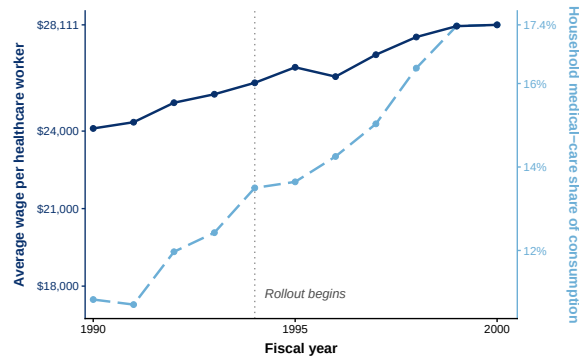
(b) Hospital birth share by ownership



(c) First stage and the rollout, by calendar year

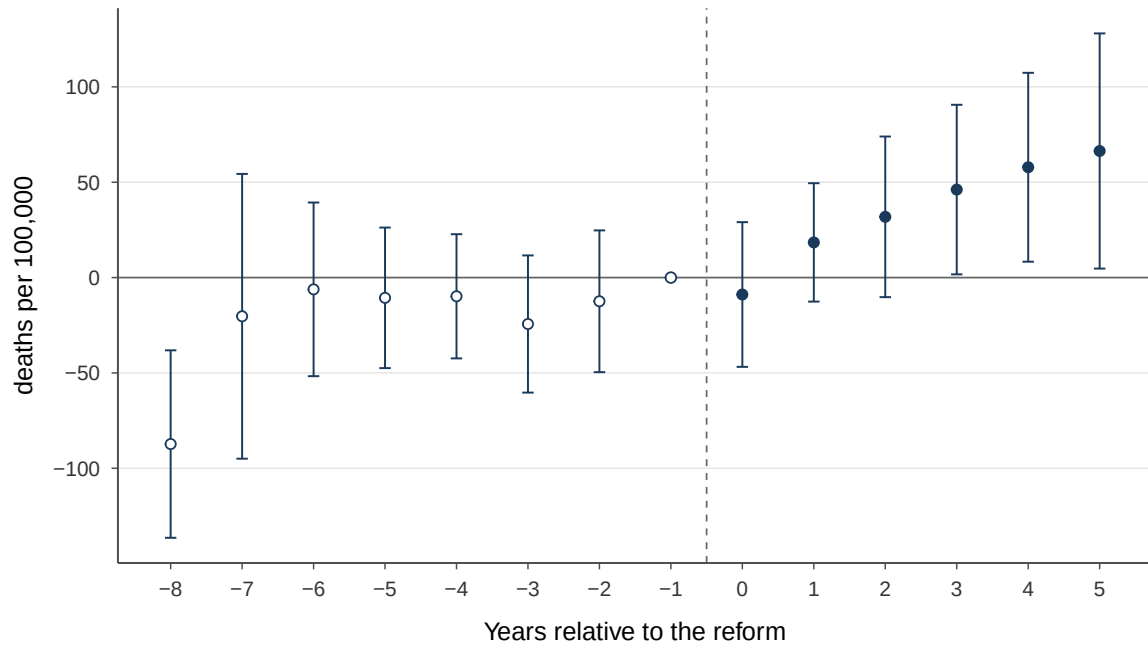


(d) Delivery spending and managed-care purchasing

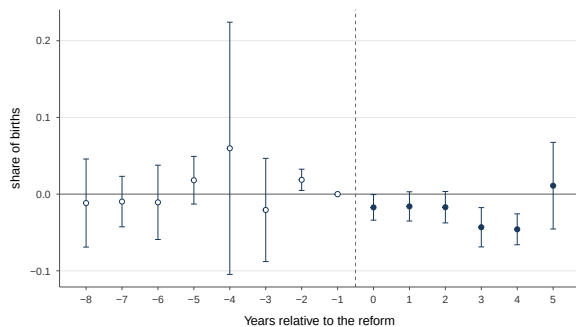


(e) Household medical spending and real wages

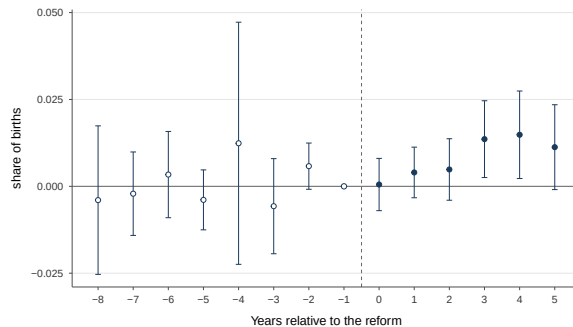
Setting



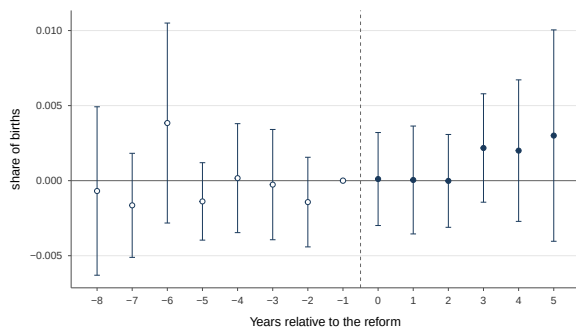
All-cause mortality



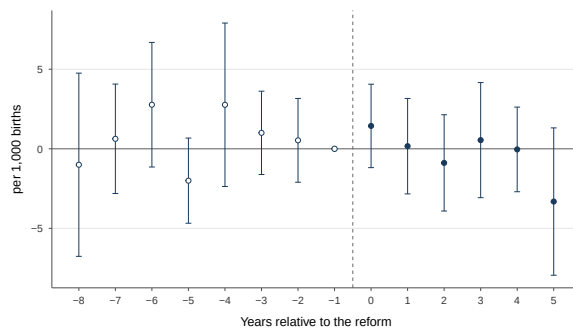
(a) Early prenatal care



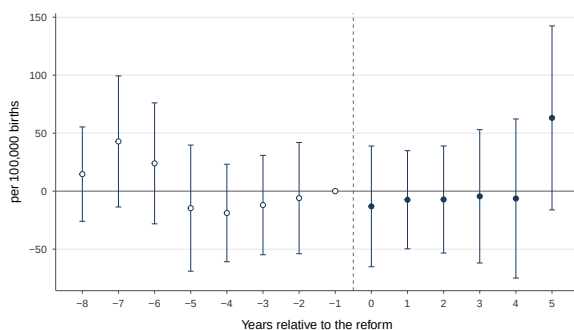
(b) Low birth weight



(c) Very low birth weight

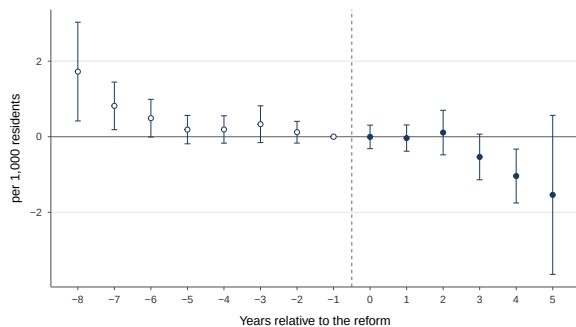


(d) Infant mortality

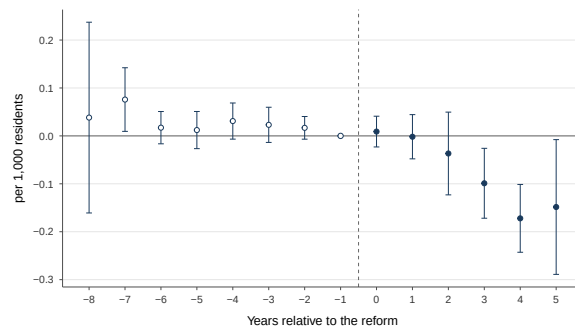


(e) Maternal mortality

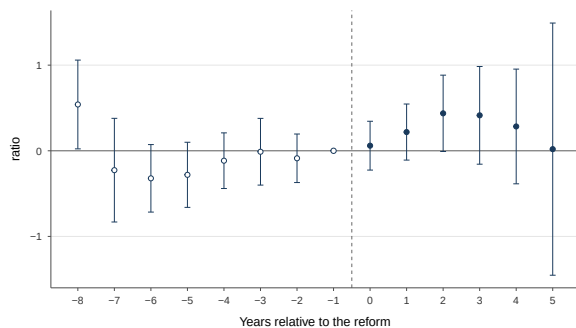
Prenatal care and prenatal-care-sensitive outcomes



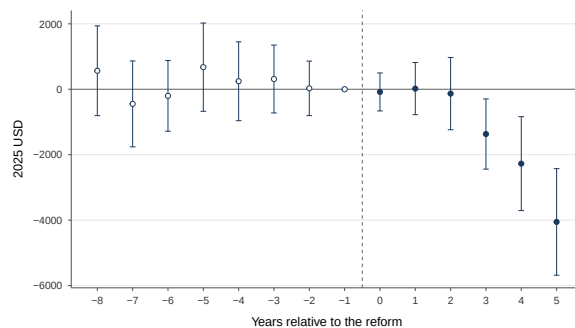
(a) Employment



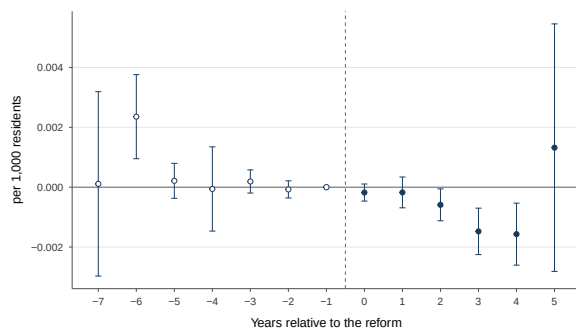
(b) Establishments



(c) Employees per establishment

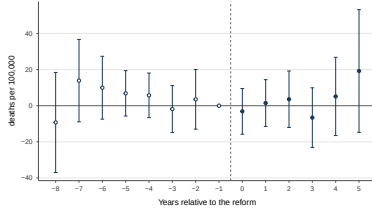


(d) Wages per worker

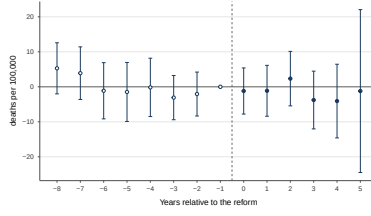


(e) Certified hospitals

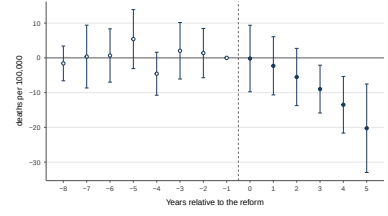
Health supply



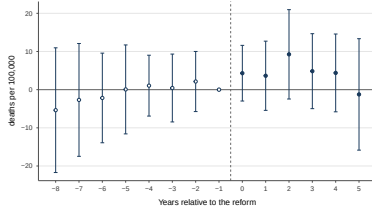
(a) Heart disease



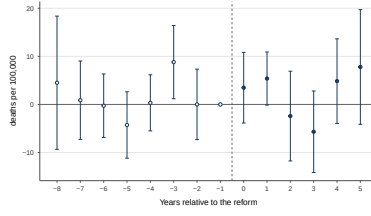
(b) Cerebrovascular disease



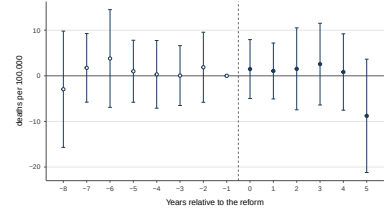
(c) Hypertension



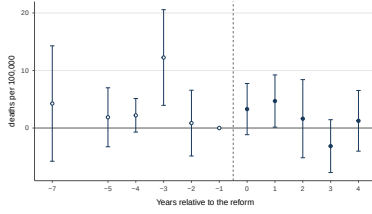
(d) Diabetes



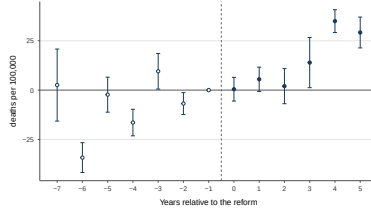
(e) Chronic pulmonary disease



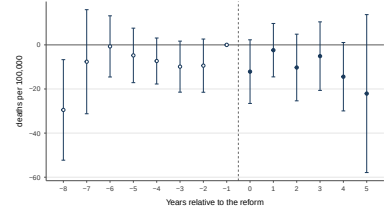
(f) Pneumonia and influenza



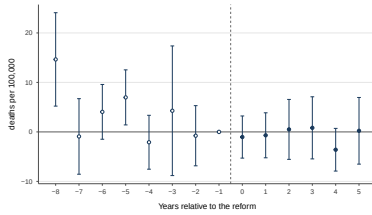
(g) Septicemia



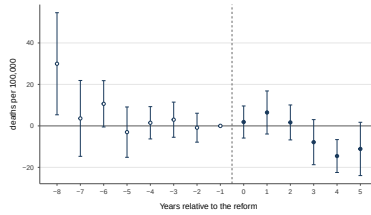
(h) AIDS



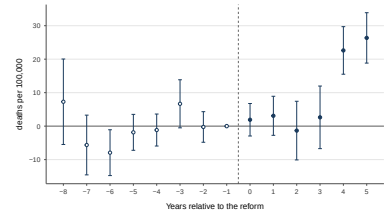
(i) Cancer



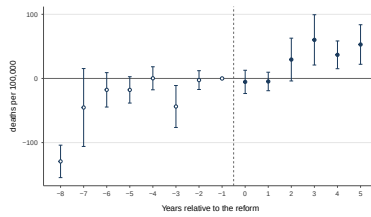
(j) Chronic liver disease



(k) Accidents

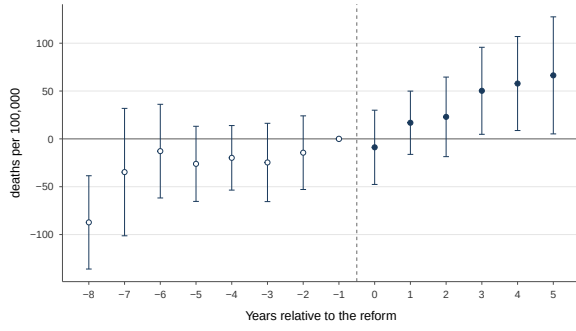


(l) Homicide

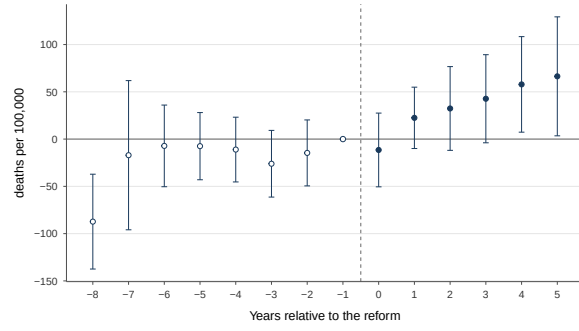


(m) Unclassified

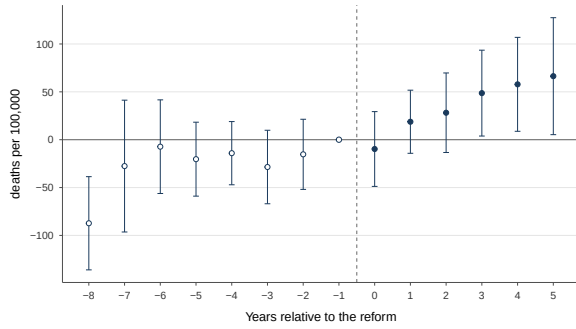
Mortality by cause



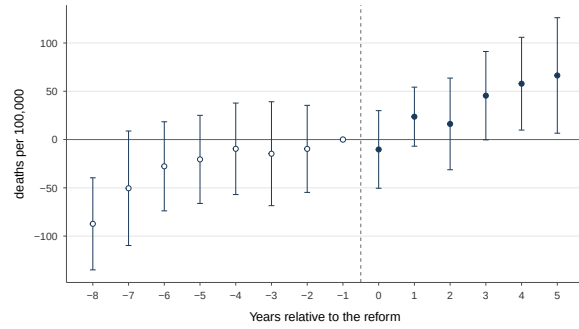
(a) Population density



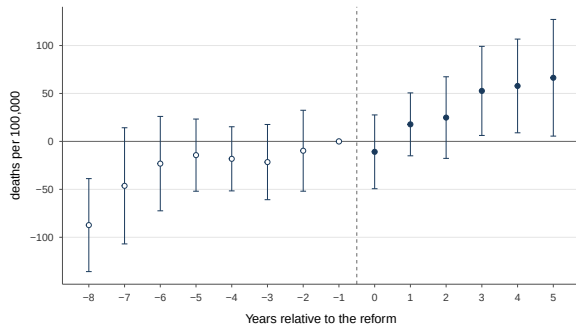
(b) Age 65 and over share



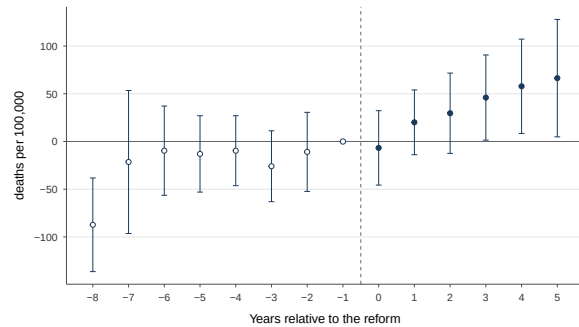
(c) Crime



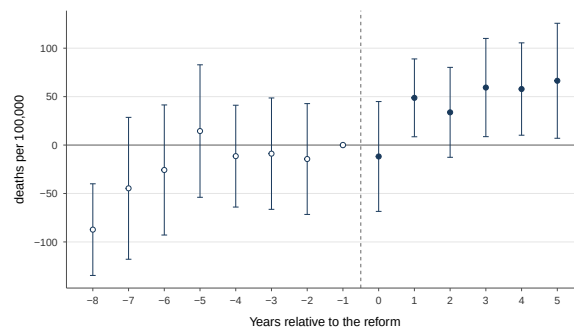
(d) Poverty rate



(e) Unemployment rate

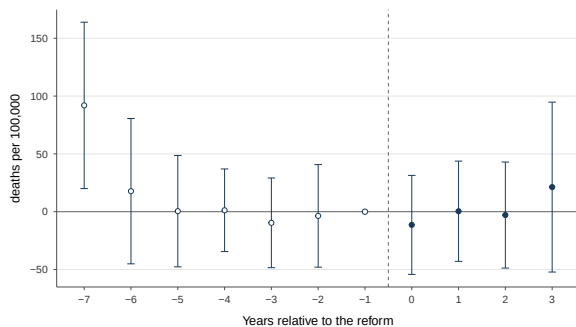


(f) Pre-reform government births

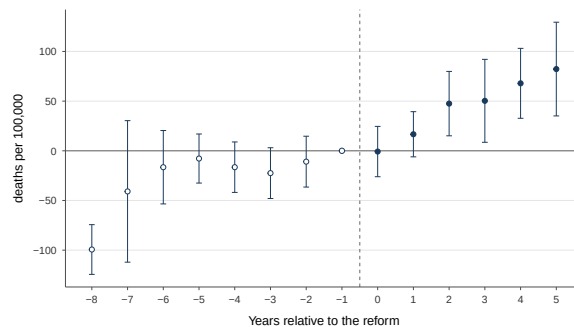


(g) All six covariates

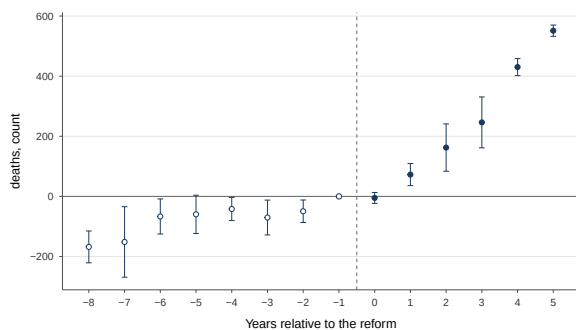
All-cause mortality under conditional trends



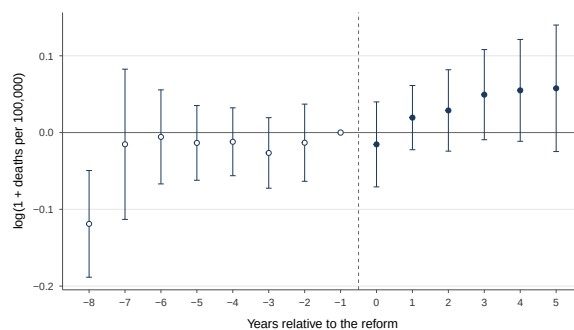
(a) Excluding San Juan



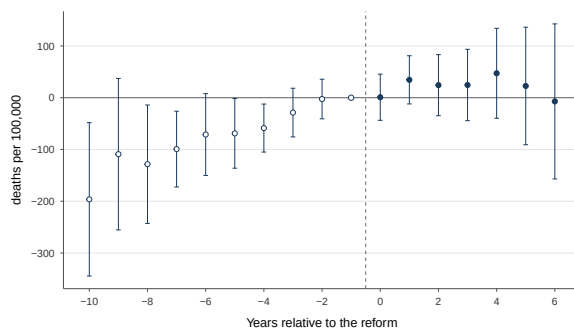
(b) Population weighted



(c) Levels

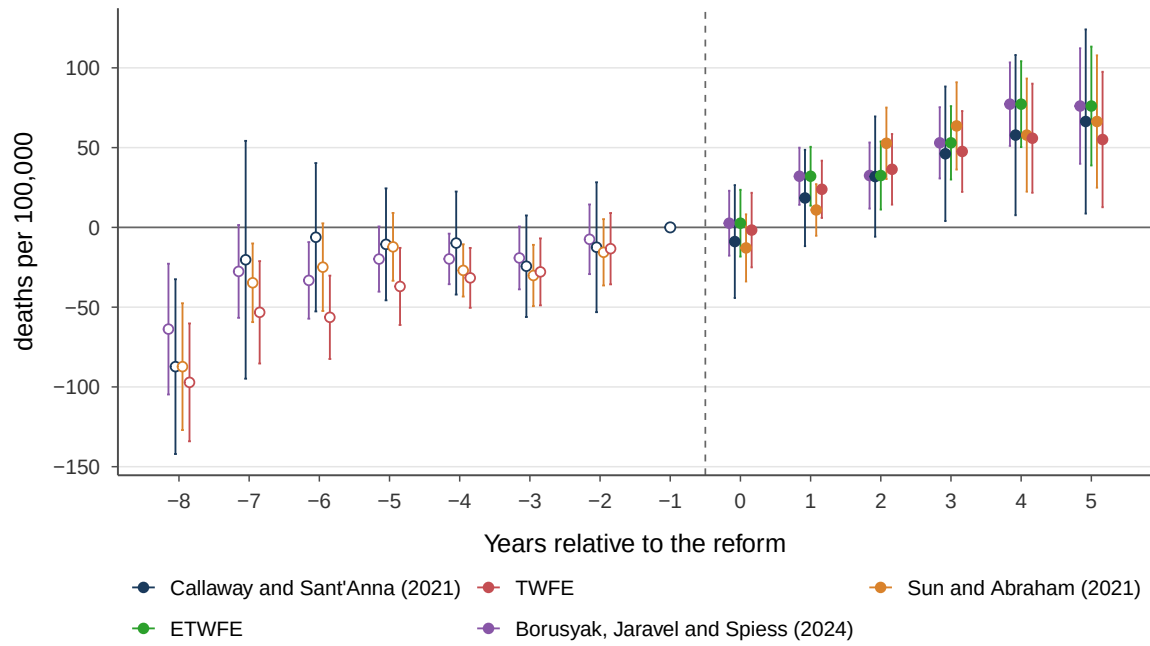


(d) Logs



(e) Never-treated, conditional

All-cause mortality under alternative specifications



Alternative estimators

Appendix: reform financing and the spending series

This appendix defines the spending series in Figure 1, distinguishing program financing from delivery costs. The island-level trends describe the reform period but do not identify its causal effects.

The reform redirected public financing toward premiums paid through the insurance administration to private insurers. Government funding continued for retained facilities and other public functions. Fiscal year 2002–2003 program financing totaled about \$2.24 billion in constant 2025 dollars: roughly 73 percent from the Commonwealth General Fund, 15 percent from federal Medicaid, 11 percent from counties, and 2 percent from the children’s health insurance program (Pan American Health Organization, 2007). Operating-budget volumes yield separate estimates of \$2.27 billion in fiscal 2002 and \$2.18 billion in fiscal 2003 (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 2004). Federal Medicaid funding was subject to a territorial cap and matching rules that differed from the income-based matching formula for states (Medicaid and CHIP Payment and Access Commission, 2020; Pillai, Pillai, Artiga, & Rudowitz, 2024).

Between fiscal years 1996 and 1999, the Puerto Rico General Fund support for health delivery and regulation, excluding the insurance administration, fell from about \$985 million to \$572 million. Own-source revenue rose from about \$382 million to \$483 million, while federal funding remained near \$650 million. Budgeted positions fell from 26,905 to 18,062, or 33 percent (Estado Libre Asociado de Puerto Rico, Oficina de Gerencia y Presupuesto, 1998).

The public-delivery series covers spending by government-operated agencies, the health services assistance fund, the Medical Center, and the Cardiovascular Center. It comes from financial statements and falls below the operating-budget totals reported elsewhere. The private-delivery series records government premiums paid to managed-care organizations, excluding household payments and privately financed insurance. Their sum covers the gov-

ernment spending represented by these two series. Household medical spending is instead a national-accounts share of personal consumption. That share rose from about 5.6 percent in 1971 to 12.4 percent in 1993 and 17.4 percent by fiscal year 2000 (Junta de Planificación de Puerto Rico, 2024).

Appendix: framework derivation

This appendix derives the effort comparisons in Section 3 and states the additional assumptions linking them to the paper’s outcomes.

Following Hart et al. (1997), a government representing society contracts with a delivery manager over a basic service bundle and a fixed payment. We treat the insurer, providers, and facility operator as one decision maker. Public or private control determines who can implement changes outside the initial contract. Eligibility, cost sharing, covered benefits, and referral arrangements are held fixed.

For service s , cost-reducing effort e_s generates production savings $c_s(e_s)$ and patient harm $b_s(e_s)$. Quality-improving effort i_s generates net benefit $\beta_s(i_s)$: patient benefits less direct production costs. With managerial effort costs normalized to $e_s + i_s$, incremental social surplus is

$$S(\mathbf{e}, \mathbf{i}) = \sum_s [c_s(e_s) - b_s(e_s) + \beta_s(i_s) - e_s - i_s]. \quad (5)$$

All components are measured in common value units. The functions vanish at zero; c_s and β_s are increasing and strictly concave, while b_s has positive, nondecreasing marginal harm. Assume $c'_s - b'_s \geq 0$ over the relevant effort range, so cost innovations remain worth implementing. Innovations are observable to the parties but noncontractible. The parties are risk neutral and split renegotiation surplus equally.

The first best maximizes S . A private manager retains cost savings without bearing the associated patient harm and receives half the net benefit of quality innovation through

renegotiation. Under public control, both innovations require approval. If the share $\lambda \in (0, 1]$ of innovation value requires the incumbent manager's participation, replacement allows the government to obtain the remainder; bargaining therefore gives the manager $\lambda/2$ of net innovation value. Assuming differentiability and interior optima,

$$\begin{aligned} c'_s(e_s^*) - b'_s(e_s^*) &= 1, & \beta'_s(i_s^*) &= 1, \\ c'_s(e_s^P) &= 1, & \frac{1}{2}\beta'_s(i_s^P) &= 1, \\ \frac{\lambda}{2} [c'_s(e_s^G) - b'_s(e_s^G)] &= 1, & \frac{\lambda}{2}\beta'_s(i_s^G) &= 1. \end{aligned} \tag{6}$$

Here $*$, P , and G denote the first best, private control, and public control. Declining marginal returns imply

$$e_s^G < e_s^* < e_s^P, \quad i_s^G \leq i_s^P < i_s^*, \tag{7}$$

with $i_s^G = i_s^P$ when $\lambda = 1$. Specifically, the public cost condition requires $c'_s - b'_s = 2/\lambda > 1$, whereas the private condition omits marginal patient harm. The quality conditions require marginal net benefits of $2/\lambda$, 2, and 1, respectively. At $\lambda = 0$, public effort is zero.

Private control thus strengthens both innovation incentives relative to public control. The balance between harm from cost reduction and benefits from quality improvement leaves patient effects ambiguous across services. Greater cost-saving effort also need not reduce total production costs, since quality improvement may require additional resources.

To connect effort to employment and capacity, suppose input j in service s satisfies

$$x_{js} = \bar{x}_{js} - \ell_{js}(e_s) + \kappa_{js}(i_s), \quad \ell'_{js} \geq 0, \quad \kappa'_{js} \geq 0. \tag{8}$$

Inputs decline under private control when the additional savings from containment exceed the additional requirements of quality improvement. Reductions in employment or establishments are therefore consistent with this mechanism but do not measure effort or efficiency.

Early prenatal care measures timely utilization, while birth outcomes and mortality re-

flect downstream health consequences. To illustrate how sustained changes in care can accumulate, let health investment $I(m_t)$ depend on effective medical inputs m_t , with $I' > 0$ and depreciation $\delta \in (0, 1)$ (M. Grossman, 1972):

$$H_{t+1} = (1 - \delta)H_t + I(m_t), \quad \Delta H_t = \Delta I \frac{1 - (1 - \delta)^t}{\delta}. \quad (9)$$

The second expression compares paths with identical initial health and a permanent investment difference ΔI beginning at $t = 0$. Reduced investment progressively lowers health relative to the counterfactual. Translating this decline into mortality requires an additional relationship between health and survival; the equation alone establishes neither a mortality lag nor a ranking of diseases. Moreover, each cause of death can depend on several services, so cause-specific mortality does not identify the quality of a corresponding service.

Appendix: data construction

This appendix records panel construction in the source order of Section 4. These checks assess data consistency and comparability, not the identifying assumptions.

The vital-statistics complement is transcribed from annual report scans. Each table is checked against printed area subtotals, island totals, and internal decompositions: government, private, and other institutional births against the place-of-birth total; and attended, unattended, and unspecified counts against institutional births. Uncertain cells are inspected in enlarged scans; missing cells are not inferred. Two exceptions remain. Because the 1990 natality table is absent from the image set, low-birth-weight and very-low-birth-weight counts use an earlier extraction, checked only to ensure each does not exceed births. In four countries, place-of-birth rows differ from Planning Board births by one or two; these discrepancies are retained. Replication documentation records the protocol, annual checks, and patches. Agreement between overlapping publications establishes consistency, not independent regis-

tration coverage.

Merging the vital-statistics complement. Source flags trace each observation in the merged Planning Board and digitized reports to its publication. Only counts are merged; rates are calculated with the panel's own denominators. Digitized records supplement early coverage of the four causes incompletely reported by the Planning Board, including observations from 1990 to 1995.

Appendix

III. Appendix Tables

Table A1: Pre-reform means by treatment cohort

Variable	1994	1995	1996	1997	1998	2000	<i>F</i>
Ambulatory health care employment per 1,000 residents	3.07	2.12	1.93	2.58	2.99	10.04	2.17
Ambulatory health care establishments per 1,000 residents	0.643	0.608	0.678	0.823	0.826	2.494	3.80
Ambulatory health care employees per establishment	4.17	3.27	2.70	2.73	3.25	4.03	1.18
Ambulatory health care wages per worker (2025 USD)	22,774	20,723	18,055	19,805	19,842	26,369	0.92
Hospitals per 1,000 residents	0.027	0.013	0.013	0.021	0.012	0.045	1.02
Hospital beds per 1,000 residents	1.58	1.06	0.91	2.27	1.67	8.73	2.22
All-cause mortality per 100,000	725.5	678.0	711.6	745.6	730.2	1,065.7	4.61
Government-hospital birth share (percent)	74.5	67.1	59.1	65.3	57.2	60.2	3.10
Private-hospital birth share (percent)	25.3	36.3	39.7	34.4	42.6	37.0	5.24
Population	36,717	29,535	35,263	51,767	62,300	443,627	23.53
Unemployment rate (percent)	19.43	18.15	21.54	20.37	14.80	10.17	3.97
Population per square mile	1,021.2	714.1	896.3	940.0	2,268.8	9,280.9	16.53
PAN recipients per 1,000 residents	455.6	529.2	458.3	466.4	358.7	313.1	7.15
Population aged 65 and over, 1990 (percent)	9.0	8.9	9.4	9.3	9.7	13.5	2.71
Families below the poverty threshold, 1990 (percent)	5,821.1	6,603.7	6,414.3	6,138.5	5,166.7	4,400.0	6.43
Labor force per 1,000 residents	305.7	287.9	345.6	301.9	340.8	352.9	4.24

Notes: Cells are county means over 1990–1993, averaged within treatment cohort. The 5 percent critical value at the median degrees of freedom is 2.34.

Table A2: Ambulatory Health Care Services employment and establishments by ownership sector, Puerto Rico (NAICS 621)

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: County Ambulatory Health Care Services (QCEW)											
Employment per 1,000 residents	2.24 (2.15)	2.60 (2.96)	2.75 (2.79)	2.90 (2.70)	3.20 (3.02)	3.27 (2.78)	3.71 (2.96)	4.18 (3.32)	4.47 (3.59)	4.68 (3.47)	4.83 (3.56)
Establishments per 1,000 residents	0.670 (0.473)	0.705 (0.479)	0.765 (0.515)	0.773 (0.513)	0.782 (0.518)	0.828 (0.564)	0.887 (0.576)	0.944 (0.601)	0.990 (0.616)	1.065 (0.640)	1.089 (0.657)
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
Panel B: County Ambulatory Health Care Services by ownership (QCEW)											
Employment	209.5 (583.0)	228.3 (603.4)	242.6 (629.9)	262.4 (689.9)	285.2 (742.9)	289.7 (729.1)	317.4 (770.3)	346.2 (784.7)	374.3 (856.3)	390.9 (873.1)	407.2 (901.3)
Total	12,362	13,472	14,313	15,484	16,827	17,091	18,726	20,426	22,084	23,064	24,025
Government total											
Private total	12,362	13,472	14,313	15,484	16,827	17,091	18,726	20,426	22,084	23,064	24,025
Establishments	56.86 (149.65)	58.54 (149.07)	63.44 (158.73)	64.41 (158.62)	65.42 (157.91)	68.14 (158.63)	71.36 (162.51)	75.68 (169.88)	79.32 (174.00)	85.25 (183.00)	87.34 (184.59)
Total	3,355	3,454	3,743	3,800	3,860	4,020	4,210	4,465	4,680	5,030	5,153
Government total											
Private total	3,355	3,454	3,743	3,800	3,860	4,020	4,210	4,465	4,680	5,030	5,153
Treated counties	0	0	0	0	9	32	37	47	58	58	59
Not-yet-treated counties	59	59	59	59	50	27	22	12	1	1	0
% of population observed	90.7	90.7	90.7	90.7	90.6	90.6	90.6	90.5	90.5	90.5	90.5
% of employment observed	97.9	97.8	97.8	97.9	97.7	97.0	96.9	96.8	96.4	96.5	96.8
% of establishments observed	97.7	97.5	97.3	97.2	97.2	96.7	96.4	96.3	96.2	96.2	96.3
Panel C: State Ambulatory Health Care Services (QCEW)											
Employment per 1,000 residents	3.58	3.88	4.09	4.37	4.67	4.74	5.24	5.68	6.11	6.32	6.52
Employment, total	12,627	13,773	14,631	15,822	17,230	17,617	19,317	21,108	22,897	23,909	24,815
Establishments per 1,000 residents	0.973	0.998	1.075	1.080	1.078	1.118	1.185	1.248	1.298	1.382	1.404
Establishments, total	3,433	3,542	3,848	3,911	3,973	4,157	4,367	4,636	4,866	5,228	5,349
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Panel D: County private healthcare sector (CBP)											
Employment per 1,000 residents	12.62 (8.79)	10.49 (8.65)	11.50 (7.98)	11.89 (9.09)	12.43 (9.38)	12.83 (9.35)	14.30 (9.05)	15.31 (9.28)	16.17 (10.00)	17.78 (11.25)	18.69 (11.27)
Employment	1,754.3 (2,551.5)	1,732.9 (2,666.1)	1,518.6 (2,484.9)	1,512.0 (2,496.0)	1,616.3 (2,712.4)	1,597.0 (2,567.8)	1,711.9 (2,940.5)	1,833.3 (3,082.3)	1,912.0 (3,077.7)	2,042.1 (3,228.8)	2,144.5 (3,256.3)
Private total	24,560	24,261	27,335	28,728	30,710	30,343	35,949	38,499	40,153	42,885	45,035
Establishments per 1,000 residents	0.813 (0.523)	0.759 (0.517)	0.799 (0.528)	0.846 (0.517)	0.878 (0.534)	0.932 (0.552)	1.308 (0.740)	1.408 (0.765)	1.445 (0.744)	1.431 (0.772)	1.474 (0.769)
Establishments	111.43 (210.12)	107.81 (203.66)	112.86 (208.67)	116.19 (203.77)	119.52 (205.67)	121.86 (205.05)	167.00 (273.55)	177.86 (286.40)	180.52 (282.67)	180.33 (280.85)	185.29 (286.01)
Private total	2,340	2,264	2,370	2,440	2,510	2,559	3,507	3,735	3,791	3,787	3,891
Treated counties	0	0	0	0	4	10	12	15	20	20	21
Not-yet-treated counties	21	21	21	21	17	11	9	6	1	1	0
% of population observed	56.8	56.8	56.8	56.8	56.6	55.6	55.5	55.3	55.1	54.9	54.7
Panel E: Individual healthcare employment (Census)											
Employment per 1,000 residents	4.66										7.33
Employment, total	16,449										27,916
% of employees with NA counties	100.0										100.0
% of population observed	100.0										100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. QCEW panels cover NAICS 621, ambulatory health care services. Percent of employment observed is the employment of the counties this panel covers in that year over the island total for the same year, and percent of establishments observed is the same quotient on establishments; health workers are employed civilians aged 16 and over in the health services group of the 1990 Census industrial classification, of which this panel counts: offices and clinics of physicians (812); offices and clinics of dentists (820); offices and clinics of chiropractors (821); offices and clinics of optometrists (822); offices and clinics of health practitioners, n.e.c. (830); health services, n.e.c. (840). These workers include the self-employed and those outside unemployment-insurance coverage. The figures are weighted estimates from the Puerto Rico five-percent microdata samples. Panel D covers the counties for which CBP published a healthcare statistic; a figure is published above 50 employees in 1990–1992 and 100 in 1993–2000. SIC 80 is the health services division as published: offices of physicians, dentists and other practitioners (801–804), nursing and personal care (805), hospitals (806), laboratories (807) and allied services

Table A3: Deaths by cause and share of all deaths by year, Puerto Rico

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)											
<i>Deaths, with share of all deaths beneath</i>											
All causes	26,148 (100.00%)	26,328 (100.00%)	27,397 (100.00%)	28,494 (100.00%)	28,444 (100.00%)	30,196 (100.00%)	29,871 (100.00%)	29,119 (100.00%)	29,990 (100.00%)	29,145 (100.00%)	28,550 (100.00%)
Diabetes	1,664 (6.36%)	1,665 (6.32%)	1,830 (6.68%)	1,870 (6.56%)	1,866 (6.56%)	2,210 (7.32%)	2,091 (7.00%)	2,097 (7.20%)	2,295 (7.65%)	2,184 (7.49%)	2,273 (7.96%)
Cancer	4,063 (15.54%)	4,278 (16.25%)	4,362 (15.92%)	4,405 (15.46%)	4,271 (15.02%)	4,619 (15.30%)	4,563 (15.28%)	4,506 (15.47%)	4,658 (15.53%)	4,429 (15.20%)	4,693 (16.44%)
Heart disease	5,910 (22.60%)	5,635 (21.40%)	5,812 (21.21%)	5,845 (20.51%)	5,783 (20.33%)	5,775 (19.13%)	5,925 (19.84%)	5,889 (20.22%)	6,038 (20.13%)	5,364 (18.40%)	5,222 (18.29%)
Cerebrovascular	1,186 (4.54%)	1,207 (4.58%)	1,275 (4.65%)	1,434 (5.03%)	1,419 (4.99%)	1,469 (4.86%)	1,465 (4.90%)	1,380 (4.74%)	1,420 (4.73%)	1,266 (4.34%)	1,166 (4.08%)
Hypertension	741 (2.83%)	948 (3.60%)	900 (3.29%)	916 (3.21%)	899 (3.16%)	1,074 (3.56%)	1,084 (3.63%)	1,190 (4.09%)	1,395 (4.65%)	1,617 (5.55%)	1,575 (5.52%)
Cirrhosis	290 (1.11%)	813 (3.09%)	804 (2.93%)	762 (2.67%)	648 (2.28%)	715 (2.37%)	670 (2.24%)	712 (2.45%)	672 (2.24%)	878 (3.01%)	820 (2.87%)
Pneumonia and influenza	1,178 (4.51%)	1,113 (4.23%)	1,192 (4.35%)	1,168 (4.10%)	1,177 (4.14%)	1,348 (4.46%)	1,174 (3.93%)	1,156 (3.97%)	1,314 (4.38%)	1,069 (3.67%)	785 (2.75%)
Septicemia	0 (0.00%)	(-)	(-)	(-)	(1.75%)	499 (1.63%)	482 (2.12%)	602 (2.07%)	599 (2.00%)	726 (2.49%)	601 (2.11%)
Chronic pulmonary	875 (3.35%)	893 (3.39%)	998 (3.64%)	1,157 (4.06%)	1,183 (4.16%)	1,145 (3.79%)	1,248 (4.18%)	1,242 (4.27%)	1,385 (4.62%)	1,240 (4.25%)	1,144 (4.01%)
AIDS	0 (0.00%)	(-)	1,407 (5.14%)	1,415 (4.97%)	1,530 (5.38%)	1,037 (3.50%)	1,474 (4.93%)	833 (2.86%)	582 (1.94%)	610 (2.09%)	659 (2.31%)
Homicide	568 (2.17%)	792 (3.01%)	838 (3.06%)	947 (3.32%)	998 (3.51%)	676 (2.34%)	928 (3.11%)	881 (3.03%)	819 (2.73%)	705 (2.42%)	700 (2.45%)
Accidents	1,503 (5.75%)	1,123 (4.27%)	1,171 (4.27%)	1,195 (4.19%)	1,287 (4.52%)	1,498 (4.96%)	1,520 (5.09%)	1,581 (5.43%)	1,387 (4.62%)	1,269 (4.35%)	1,321 (4.63%)
Unclassified	8,170 (31.25%)	7,861 (29.86%)	6,823 (24.90%)	7,380 (25.90%)	6,884 (24.20%)	8,151 (26.99%)	7,095 (23.75%)	7,050 (24.21%)	7,426 (24.76%)	7,788 (26.72%)	7,591 (26.59%)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: Entries are TOTAL deaths in Puerto Rico, with that cause's share of all deaths in parentheses beneath. Both are island figures, not county averages. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified is all-cause deaths less the causes listed, so the shares sum to 100; where a cause is unreported its deaths fall into that row; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative. Sample: unbalanced, every county the panel observes, with no balance restriction.

Table A4: Mortality and birth-margin outcomes by year, Puerto Rico counties

Variable	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Panel A: Population, mortality and births (Puerto Rico Planning Board)											
<i>Mortality</i>											
All-cause mortality per 100,000	703.5 (116.7)	696.9 (103.9)	720.2 (103.9)	744.5 (108.0)	728.4 (120.9)	768.4 (132.4)	763.5 (126.5)	759.8 (114.1)	767.2 (114.3)	734.7 (109.6)	718.3 (113.2)
Diabetes mortality per 100,000	46.9 (23.4)	46.0 (17.4)	54.3 (22.0)	52.2 (23.6)	50.7 (19.2)	59.9 (22.8)	57.0 (24.1)	57.6 (21.1)	62.6 (23.2)	58.6 (19.9)	59.5 (18.9)
Cancer mortality per 100,000	108.1 (27.6)	112.2 (27.7)	116.7 (25.0)	118.8 (27.4)	111.2 (34.6)	118.6 (27.2)	118.1 (29.8)	119.9 (29.9)	120.0 (24.1)	108.3 (30.5)	117.7 (23.8)
Heart disease mortality per 100,000	163.9 (36.6)	155.1 (33.6)	156.8 (38.8)	154.7 (30.0)	151.7 (31.6)	153.4 (41.0)	153.8 (37.4)	153.7 (34.3)	153.9 (40.7)	139.0 (43.2)	133.3 (38.4)
Cerebrovascular mortality per 100,000	34.7 (18.8)	34.8 (12.4)	36.0 (12.5)	39.5 (11.5)	37.7 (15.8)	40.0 (13.2)	39.5 (12.6)	37.3 (15.4)	36.0 (13.8)	34.5 (17.3)	30.4 (12.4)
Hypertension mortality per 100,000	21.2 (9.3)	29.3 (18.9)	26.7 (11.7)	26.0 (16.8)	25.7 (12.2)	32.9 (27.9)	30.4 (14.2)	33.6 (11.7)	36.6 (12.1)	40.5 (15.1)	40.0 (13.3)
Cirrhosis mortality per 100,000	7.6 (11.2)	23.6 (11.5)	22.9 (31.5)	21.6 (9.6)	17.3 (7.8)	18.4 (8.2)	19.2 (10.8)	19.5 (11.8)	18.0 (10.2)	23.3 (9.9)	20.7 (10.5)
Pneumonia and influenza mortality per 100,000	29.8 (14.9)	28.2 (12.5)	29.4 (12.8)	30.0 (13.2)	28.1 (14.4)	31.2 (11.2)	28.7 (11.3)	28.2 (12.4)	32.5 (15.5)	25.0 (10.9)	20.1 (10.3)
Septicemia mortality per 100,000	0.0 (0.0)	(-)	(-)	(-)	12.4 (7.4)	14.8 (7.3)	16.2 (9.1)	15.5 (9.1)	14.9 (7.6)	17.6 (8.3)	14.3 (7.2)
Chronic pulmonary mortality per 100,000	23.9 (12.5)	25.9 (12.8)	29.0 (14.8)	32.1 (16.6)	33.7 (14.5)	36.3 (18.3)	34.0 (14.4)	32.2 (15.8)	37.3 (16.2)	34.4 (16.5)	31.2 (15.9)
AIDS mortality per 100,000	0.0 (0.0)	(-)	26.6 (16.7)	27.5 (17.4)	29.4 (19.0)	27.7 (15.4)	30.2 (17.0)	17.3 (12.2)	12.0 (9.4)	12.9 (9.1)	13.4 (9.2)
Homicide mortality per 100,000	9.7 (9.7)	13.3 (13.7)	13.0 (13.4)	16.1 (15.1)	17.6 (18.1)	18.0 (14.7)	15.9 (15.1)	17.7 (15.4)	15.5 (16.1)	12.9 (11.1)	14.3 (13.2)
Accident mortality per 100,000	43.3 (22.7)	32.8 (15.2)	33.7 (13.7)	32.7 (12.5)	35.4 (15.7)	41.5 (23.7)	39.4 (14.2)	40.8 (15.1)	37.6 (15.9)	30.9 (13.1)	31.7 (12.3)
Unclassified mortality per 100,000	214.5 (61.6)	195.7 (43.3)	175.7 (40.9)	193.3 (39.1)	177.3 (37.1)	181.1 (56.1)	181.0 (38.0)	186.5 (39.0)	190.3 (38.8)	196.9 (38.5)	191.7 (44.2)
<i>Birth outcomes</i>											
Infant mortality per 1,000 live births	12.75 (6.09)	13.78 (11.74)	12.39 (4.85)	13.13 (5.80)	11.55 (5.26)	14.08 (7.44)	9.91 (4.90)	12.11 (6.42)	10.48 (5.02)	10.10 (4.93)	9.91 (5.60)
Low birth weight share (percent)	9.28 (1.62)	9.98 (7.51)	(-)	9.65 (1.64)	9.86 (1.44)	10.20 (1.76)	10.49 (1.90)	11.07 (2.33)	10.98 (2.24)	11.36 (2.46)	2.46 (1.34)
Early prenatal care share (percent)	71.9 (6.7)	77.8 (34.1)	(-)	77.7 (6.7)	78.5 (7.2)	77.9 (6.2)	78.6 (5.9)	79.0 (6.8)	79.7 (6.5)	79.7 (6.4)	78.6 (8.7)
Treated counties	0	0	0	0	18	45	52	65	77	77	78
Not-yet-treated counties	78	78	78	78	60	33	26	13	1	1	0
% of population observed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: County means with standard deviations in parentheses. Percent of population observed is the population of the counties this panel covers in that year, divided by the population of every county in that year. Unclassified mortality is all-cause deaths less every cause identified, expressed per 100,000 residents; in two county-years the published cause counts exceed published all-cause deaths, by 15 in Hatillo in 1992 and by 3 in Las Marias in 1995, and the residual is floored at zero there rather than carried negative.

Table A5: Two-Way Fixed Effects with Control Variables: Contemporaneous Covariates

	(1)	(2)	(3)	(4)	(5)	(6)
<i>All-cause mortality</i>						
<i>Contemporaneous covariates, in levels</i>						
ATT (per 100,000)	-1.283 (10.20)	0.7695 (10.34)	-1.549 (10.15)	-1.756 (9.907)	3.619 (10.44)	3.413 (9.836)
Pre-treatment mean	724.2	724.2	724.2	724.2	724.2	724.2
% of pre-treatment mean	-0.2%	0.1%	-0.2%	-0.2%	0.5%	0.5%
SE relative to unconditional		1.013	0.9945	0.9709	1.024	0.9640
Counties	78	78	78	78	78	78
<i>Covariates</i>						
Population density		✓				✓
Unemployment rate			✓			✓
Crime				✓		✓
PAN recipient share					✓	✓
Estimated through	1999	1999	1999	1999	1999	1999
Parameters	.	2	2	2	2	5
Clusters	78	78	78	78	78	78

Notes: Two-way fixed effects, ordinary least squares. Each covariate enters in levels at its value in that year. Standard errors in parentheses: for TWFE and TWFE-C, clustered on the county. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Each covariate is measured in the same year as the outcome, so in a treated county it is measured after that county's reform year. Each control variable is one number per county and year. annual county values: population density, unemployment rate, crime and pan recipient share. The columns differ in what they condition on and rest on the same counties and years. Two-way fixed effects under staggered adoption weights 32.3% of treated county-years negatively; the negative weights are 16.3% of the positive ones in absolute value. Density is residents per square mile in that year; unemployment is the county annual rate in that year; crime is total reported offences per 1,000 residents in that year, murders included, on the definition the baseline crime control also uses; the PAN recipient share is residents receiving the Programa de Asistencia Nutricional over county population in that year. Dataset is balanced.

Table A6: Alternative Specifications: Mortality by Cause and Prenatal-Care-Sensitive Outcomes

	(1) Baseline	(2) Excluding San Juan	(3) Population weighted	(4) Levels	(5) Logs	(6) Never-treated
<i>Panel C. Mortality</i>						
<i>All-cause mortality</i>						
All causes (per 100,000; pre-treatment mean 724.2)	26.96*** (10.32)	-2.600 (12.65)	32.27*** (8.408)	173.2*** (9.233)	0.0247* (0.0145)	23.15 (20.54)
% of pre-treatment mean	3.7%	-0.4%	4.5%	54.2%	2.5%	3.2%
<i>Cardiovascular and metabolic causes</i>						
Heart disease (per 100,000; pre-treatment mean 156.6)	1.009 (3.980)	-4.693 (4.743)	1.703 (3.725)	38.00*** (2.145)	-0.0211 (0.0327)	36.37*** (13.09)
% of pre-treatment mean	0.6%	-3.0%	1.1%	56.4%	-2.1%	23.1%
Cerebrovascular (per 100,000; pre-treatment mean 36.72)	-1.285 (2.306)	0.7241 (2.652)	-1.962 (1.653)	3.025*** (1.025)	0.0038 (0.0700)	5.481 (9.859)
% of pre-treatment mean	-3.5%	2.0%	-5.3%	19.2%	0.4%	14.9%
Hypertension (per 100,000; pre-treatment mean 26.68)	-5.943*** (2.221)	1.136 (2.862)	-6.500*** (1.231)	-46.11*** (1.329)	-0.0707 (0.0859)	
% of pre-treatment mean	-22.3%	4.3%	-24.4%	-412.2%	-6.8%	
Diabetes (per 100,000; pre-treatment mean 51.55)	4.913** (2.409)	5.095 (3.108)	4.444** (1.803)	-2.263 (1.469)	0.1079** (0.0479)	-10.45 (9.981)
% of pre-treatment mean	9.5%	9.9%	8.6%	-10.3%	11.4%	-20.2%
<i>Respiratory and infectious causes</i>						
Pneumonia and influenza (per 100,000; pre-treatment mean 29.44)	0.9216 (2.363)	0.3048 (2.959)	1.645 (1.770)	6.417*** (1.658)	0.0068 (0.1143)	
% of pre-treatment mean	3.1%	1.0%	5.6%	47.8%	0.7%	
<i>Cancer and chronic liver disease</i>						
Cancer (per 100,000; pre-treatment mean 114.4)	-9.277** (4.245)	-11.35** (5.403)	-5.490** (2.792)	3.878* (2.127)	-0.0874* (0.0507)	20.16*** (5.241)
% of pre-treatment mean	-8.1%	-9.9%	-4.8%	7.6%	-8.4%	17.5%
Cirrhosis (per 100,000; pre-treatment mean 18.64)	-0.6440 (1.245)	0.1310 (1.806)	-0.7293 (0.9615)	-4.479*** (0.8057)	0.0043 (0.1131)	107.2*** (14.10)
% of pre-treatment mean	-3.5%	0.7%	-3.9%	-55.6%	0.4%	573.9%
<i>External causes</i>						
Accidents (per 100,000; pre-treatment mean 36.02)	-1.533 (2.433)	1.804 (2.832)	-2.554 (2.800)	-4.770 (4.360)	-0.0903 (0.0752)	6.410 (4.488)
% of pre-treatment mean	-4.3%	5.0%	-7.1%	-31.3%	-8.6%	17.7%
<i>Panel F. Unclassified</i>						
Unclassified (per 100,000; pre-treatment mean 254.2)	38.79*** (6.722)	4.252 (8.877)	41.72*** (7.432)	179.5*** (11.48)	0.0754** (0.0354)	
% of pre-treatment mean	15.3%	1.7%	16.4%	155.4%	7.8%	
<i>Panel D. Prenatal-care-sensitive outcomes</i>						
Infant mortality (per 1,000 births; pre-treatment mean 12.56)	0.0985 (0.7941)	0.6985 (1.077)	-0.5230 (0.6708)	6.531*** (1.204)	0.0170 (0.1013)	
% of pre-treatment mean	0.8%	5.6%	-4.2%	66.8%	1.7%	
Maternal mortality (per 100,000 births; pre-treatment mean 19.27)	-4.291 (22.43)	-19.70 (25.65)	8.345 (7.469)	0.7297*** (0.0773)	1.362*** (0.2559)	
% of pre-treatment mean	-22.3%	-102.2%	43.3%	578.5%	290.4%	
Treated counties	78	77	78	78	78	78
Never-treated counties						147

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico counties, 1990 to 1999. The control group is the not-yet-treated counties. Every column is the Callaway and Sant’Anna estimator; each varies one feature of the DESIGN rather than the estimator. San Juan is the sole comparison unit in 1998 and 1999, so the column excluding it shows what survives without it. The population-weighted column estimates the effect on the average resident rather than the average county, at the primary scale, so it is read against column (1) rather than against Levels. A positive log estimate beside a negative level estimate is distributional heterogeneity, not a contradiction. The last column replaces the comparison group rather than the estimation, and is conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census), matching the specification the never-treated tables report. Non-MMC states had not adopted Medicaid managed care by the horizon. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Treated counties are reported once per source, since every outcome from a source rests on the same roster; they differ across sources because the employment, provider and vital-statistics panels do not cover the same counties. Dataset is balanced.

Table A7: Never-Treated Control Groups, Conditional

	(1) Non-MMC	(2) Non-HMO	(3) Non-MMC, high gov %	(4) Non-HMO, high gov %
<i>All-cause mortality</i>				
ATT	23.15 (20.54)	21.57 (21.67)	23.15 (20.54)	21.57 (21.67)
Pre-treatment mean	731.8	731.8	731.8	731.8
ATT as % of pre-treatment mean	3.2%	2.9%	3.2%	2.9%
Treated counties	78	78	78	78
Never-treated counties	282	353	282	353

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The outcome is measured per 100,000. Puerto Rico against never-treated mainland counties, through 2000. Each column is a rule for admitting never-treated counties; all are on the mainland, since every Puerto Rico county is treated by 2000. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon; high government share means government facilities exceed half of Medicare-certified providers, matching Puerto Rico’s pre-reform system. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Non-MMC, high government share: 76 counties in 5 states, Mississippi (32), Illinois (19), Wyoming (12), South Carolina (11), Alaska (2). Non-HMO, high government share: 82 counties in 7 states, Idaho (19), Alabama (18), Louisiana (15), Wyoming (12), Arkansas (8), South Dakota (8), Alaska (2). Never-treated mortality comes from CDC WONDER. This horizon crosses the ICD-9 to ICD-10 transition, and Puerto Rico and the mainland cross it under different coding systems, so these estimates carry a source break that a year effect does not remove. Read them as a sensitivity on horizon length, not as a longer-run estimate of the quantity in the 2000 table. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census): parallel trends are assumed among counties sharing those baseline values rather than across all of them. The island poverty rate is families below the Census threshold over all families; the donor rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise; the estimator per cell is recorded in the digest. Dataset is balanced.

Table A8: Never-Treated Control Groups: Mortality by Cause, Conditional

	(1) Non-MMC	(2) Non-HMO	(3) Non-MMC, high gov %	(4) Non-HMO, high gov %
<i>Panel C. Mortality</i>				
<i>All-cause mortality</i>				
All causes (per 100,000; pre-treatment mean 731.8; 78 treated counties)	23.15 (20.54)	21.57 (21.67)	23.15 (20.54)	21.57 (21.67)
% of pre-treatment mean	3.2%	2.9%	3.2%	2.9%
Never-treated counties	282	353	282	353
<i>Cardiovascular and metabolic causes</i>				
Heart disease (per 100,000; pre-treatment mean 157.7; 78 treated counties)	36.37*** (13.09)	38.68*** (12.40)	36.37*** (13.09)	38.68*** (12.40)
% of pre-treatment mean	23.1%	24.5%	23.1%	24.5%
Never-treated counties	255	273	255	273
Cerebrovascular (per 100,000; pre-treatment mean 36.84; 78 treated counties)	5.481 (9.859)	-6.707 (6.453)	5.481 (9.859)	-6.707 (6.453)
% of pre-treatment mean	14.9%	-18.2%	14.9%	-18.2%
Never-treated counties	148	146	148	146
Diabetes (per 100,000; pre-treatment mean 51.73; 78 treated counties)	-10.45 (9.981)	-10.11 (6.793)	-10.45 (9.981)	-10.11 (6.793)
% of pre-treatment mean	-20.2%	-19.5%	-20.2%	-19.5%
Never-treated counties	65	56	65	56
<i>Respiratory and infectious causes</i>				
Pneumonia and influenza (per 100,000; pre-treatment mean 29.88; 78 treated counties)		-4.159 (4.960)		-4.159 (4.960)
% of pre-treatment mean		-13.9%		-13.9%
Never-treated counties		51		51
<i>Cancer and chronic liver disease</i>				
Cancer (per 100,000; pre-treatment mean 115.3; 78 treated counties)	20.16*** (5.241)	25.75*** (5.650)	20.16*** (5.241)	25.75*** (5.650)
% of pre-treatment mean	17.5%	22.3%	17.5%	22.3%
Never-treated counties	188	194	188	194
Cirrhosis (per 100,000; pre-treatment mean 18.68; 78 treated counties)	107.2*** (14.10)	106.5*** (14.49)	107.2*** (14.10)	106.5*** (14.49)
% of pre-treatment mean	573.9%	569.9%	573.9%	569.9%
Never-treated counties	255	273	255	273
<i>External causes</i>				
Accidents (per 100,000; pre-treatment mean 36.27; 78 treated counties)	6.410 (4.488)	0.9650 (7.151)	6.410 (4.488)	0.9650 (7.151)
% of pre-treatment mean	17.7%	2.7%	17.7%	2.7%
Never-treated counties	147	139	147	139

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico against never-treated mainland counties, through 2000. Each column is a rule for admitting never-treated counties; all are on the mainland, since every Puerto Rico county is treated by 2000. The pre-treatment mean and treated-county count are in each row label, being identical across columns. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon; high government share means government facilities exceed half of Medicare-certified providers, matching Puerto Rico’s pre-reform system. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Non-MMC, high government share: 76 counties in 5 states, Mississippi (32), Illinois (19), Wyoming (12), South Carolina (11), Alaska (2). Non-HMO, high government share: 82 counties in 7 states, Idaho (19), Alabama (18), Louisiana (15), Wyoming (12), Arkansas (8), South Dakota (8), Alaska (2). Never-treated mortality comes from CDC WONDER. This horizon crosses the ICD-9 to ICD-10 transition, and Puerto Rico and the mainland cross it under different coding systems, so these estimates carry a source break that a year effect does not remove. Read them as a sensitivity on horizon length, not as a longer-run estimate of the quantity in the 2000 table. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census): parallel trends are assumed among counties sharing those baseline values rather than across all of them. The island poverty rate is families below the Census threshold over all families; the donor rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise; the estimator per cell is recorded in the digest. Dataset is balanced.

Table A9: Never-Treated Control Groups (until 2003), Conditional

	(1) Non-MMC	(2) Non-HMO	(3) Non-MMC, high gov %	(4) Non-HMO, high gov %
<i>All-cause mortality</i>				
ATT	16.22 (19.39)	14.93 (27.23)	16.22 (19.39)	14.93 (27.23)
Pre-treatment mean	731.8	731.8	731.8	731.8
ATT as % of pre-treatment mean	2.2%	2.0%	2.2%	2.0%
Treated counties	78	78	78	78
Never-treated counties	282	353	282	353

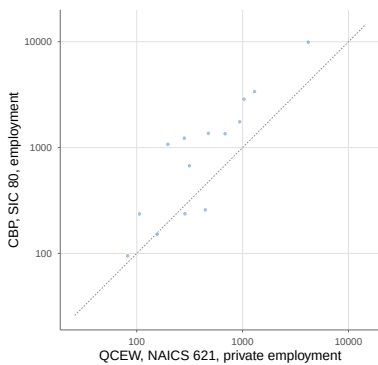
Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The outcome is measured per 100,000. Puerto Rico against never-treated mainland counties, through 2003. Each column is a rule for admitting never-treated counties; all are on the mainland, since every Puerto Rico county is treated by 2000. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon; high government share means government facilities exceed half of Medicare-certified providers, matching Puerto Rico’s pre-reform system. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Non-MMC, high government share: 76 counties in 5 states, Mississippi (32), Illinois (19), Wyoming (12), South Carolina (11), Alaska (2). Non-HMO, high government share: 82 counties in 7 states, Idaho (19), Alabama (18), Louisiana (15), Wyoming (12), Arkansas (8), South Dakota (8), Alaska (2). Never-treated mortality comes from CDC WONDER. This horizon crosses the ICD-9 to ICD-10 transition, and Puerto Rico and the mainland cross it under different coding systems, so these estimates carry a source break that a year effect does not remove. Read them as a sensitivity on horizon length, not as a longer-run estimate of the quantity in the 2000 table. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census): parallel trends are assumed among counties sharing those baseline values rather than across all of them. The island poverty rate is families below the Census threshold over all families; the donor rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise; the estimator per cell is recorded in the digest. Dataset is balanced.

Table A10: Never-Treated Control Groups: Mortality by Cause (until 2003), Conditional

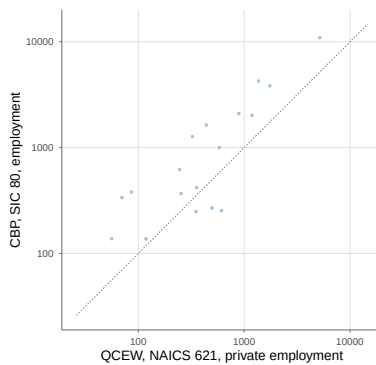
	(1) Non-MMC	(2) Non-HMO	(3) Non-MMC, high gov %	(4) Non-HMO, high gov %
<i>Panel C. Mortality</i>				
<i>All-cause mortality</i>				
All causes (per 100,000; pre-treatment mean 731.8; 78 treated counties)	16.22 (19.39)	14.93 (27.23)	16.22 (19.39)	14.93 (27.23)
% of pre-treatment mean	2.2%	2.0%	2.2%	2.0%
Never-treated counties	282	353	282	353
<i>Cardiovascular and metabolic causes</i>				
Heart disease (per 100,000; pre-treatment mean 157.7; 78 treated counties)	69.55*** (13.88)	59.73*** (14.71)	69.55*** (13.88)	59.73*** (14.71)
% of pre-treatment mean	44.1%	37.9%	44.1%	37.9%
Never-treated counties	255	273	255	273
Cerebrovascular (per 100,000; pre-treatment mean 36.84; 78 treated counties)	-0.4586 (11.03)	-9.508 (7.093)	-0.4586 (11.03)	-9.508 (7.093)
% of pre-treatment mean	-1.2%	-25.8%	-1.2%	-25.8%
Never-treated counties	148	146	148	146
Diabetes (per 100,000; pre-treatment mean 51.73; 78 treated counties)	-10.00 (8.037)	-11.18 (7.140)	-10.00 (8.037)	-11.18 (7.140)
% of pre-treatment mean	-19.3%	-21.6%	-19.3%	-21.6%
Never-treated counties	66	56	66	56
<i>Respiratory and infectious causes</i>				
Pneumonia and influenza (per 100,000; pre-treatment mean 29.88; 78 treated counties)		-7.954* (4.797)		-7.954* (4.797)
% of pre-treatment mean		-26.6%		-26.6%
Never-treated counties		53		53
<i>Cancer and chronic liver disease</i>				
Cancer (per 100,000; pre-treatment mean 115.3; 78 treated counties)	31.46*** (5.493)	37.02*** (6.269)	31.46*** (5.493)	37.02*** (6.269)
% of pre-treatment mean	27.3%	32.1%	27.3%	32.1%
Never-treated counties	188	194	188	194
Cirrhosis (per 100,000; pre-treatment mean 18.68; 78 treated counties)	183.8*** (16.31)	164.9*** (19.09)	183.8*** (16.31)	164.9*** (19.09)
% of pre-treatment mean	983.5%	882.5%	983.5%	882.5%
Never-treated counties	255	273	255	273
<i>External causes</i>				
Accidents (per 100,000; pre-treatment mean 36.27; 78 treated counties)	11.70*** (4.138)	1.833 (9.110)	11.70*** (4.138)	1.833 (9.110)
% of pre-treatment mean	32.3%	5.1%	32.3%	5.1%
Never-treated counties	149	140	149	140

Notes: Callaway and Sant’Anna (2021). Standard errors in parentheses: county-clustered multiplier bootstrap, 1,000 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Puerto Rico against never-treated mainland counties, through 2003. Each column is a rule for admitting never-treated counties; all are on the mainland, since every Puerto Rico county is treated by 2000. The pre-treatment mean and treated-county count are in each row label, being identical across columns. Non-MMC states had not adopted Medicaid managed care by the horizon; non-HMO states operated no Medicaid HMO by the horizon; high government share means government facilities exceed half of Medicare-certified providers, matching Puerto Rico’s pre-reform system. Non-MMC: 302 counties in 6 states, Illinois (102), Mississippi (82), South Carolina (46), Alaska (39), Wyoming (23), New Hampshire (10). Non-HMO: 378 counties in 7 states, Arkansas (75), Alabama (67), South Dakota (66), Louisiana (64), Idaho (44), Alaska (39), Wyoming (23). Non-MMC, high government share: 76 counties in 5 states, Mississippi (32), Illinois (19), Wyoming (12), South Carolina (11), Alaska (2). Non-HMO, high government share: 82 counties in 7 states, Idaho (19), Alabama (18), Louisiana (15), Wyoming (12), Arkansas (8), South Dakota (8), Alaska (2). Never-treated mortality comes from CDC WONDER. This horizon crosses the ICD-9 to ICD-10 transition, and Puerto Rico and the mainland cross it under different coding systems, so these estimates carry a source break that a year effect does not remove. Read them as a sensitivity on horizon length, not as a longer-run estimate of the quantity in the 2000 table. Conditional on population density (1990 to 1993 mean) and poverty rate (1990 Census): parallel trends are assumed among counties sharing those baseline values rather than across all of them. The island poverty rate is families below the Census threshold over all families; the donor rate is persons for whom poverty status is determined. Estimated doubly-robust where that fit converged and by outcome regression otherwise; the estimator per cell is recorded in the digest. Dataset is balanced.

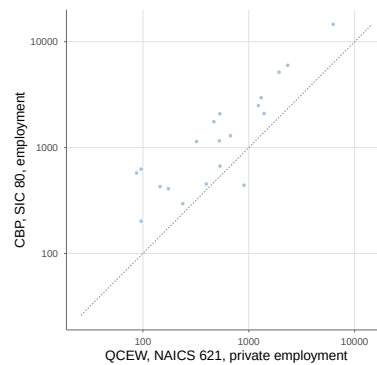
IV. Appendix Figures



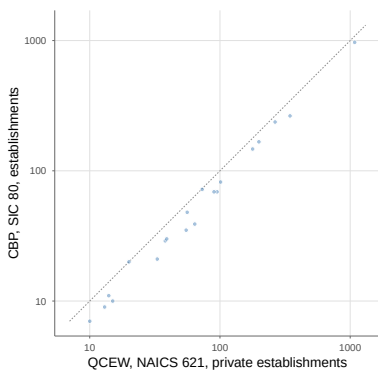
(a) 1990: CBP – employment



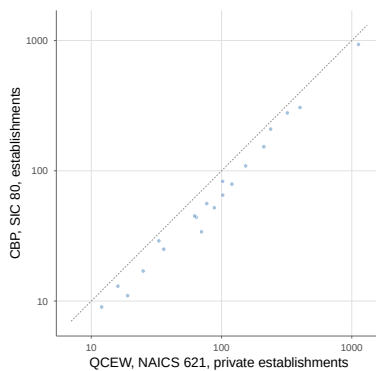
(b) 1995: CBP – employment



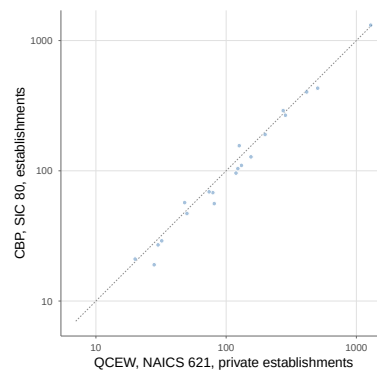
(c) 2000: CBP – employment



(d) 1990: CBP – establish-
ments



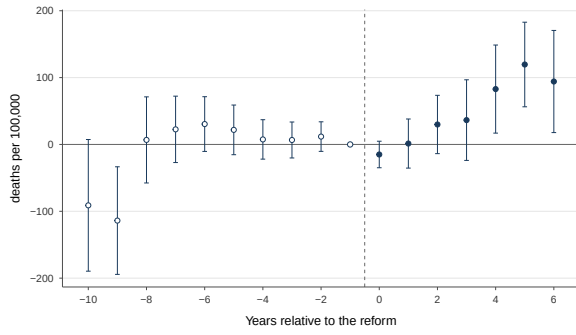
(e) 1995: CBP – establish-
ments



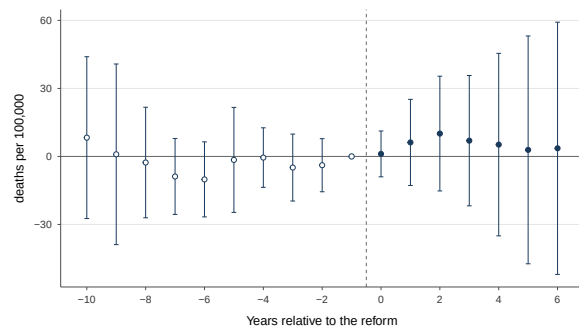
(f) 2000: CBP – establish-
ments

QCEW compared to CBP

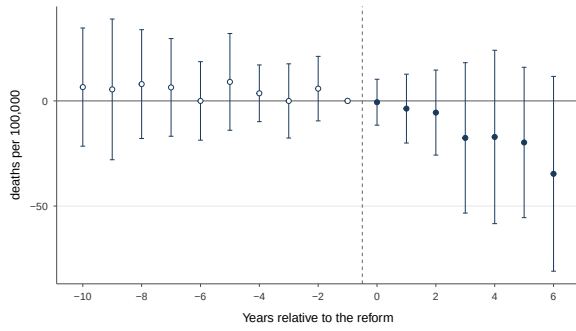
Each point is one county in the year the panel names, over the counties both sources publish a positive figure for. QCEW is NAICS 621, ambulatory health care services, restricted to private ownership; CBP is SIC 80, all health services, so it also carries hospitals and nursing and residential care. CBP counts a single reference period and QCEW an annual average. Counties drawn: employment 14 to 20 per year, establishments 20.



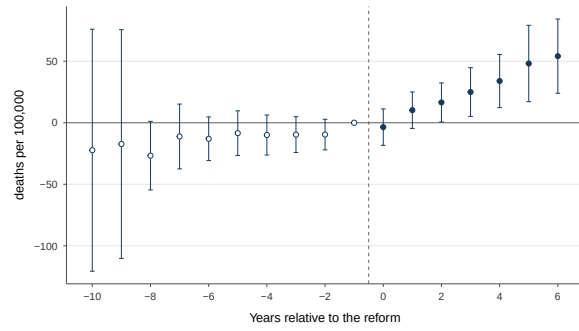
(a) Heart disease



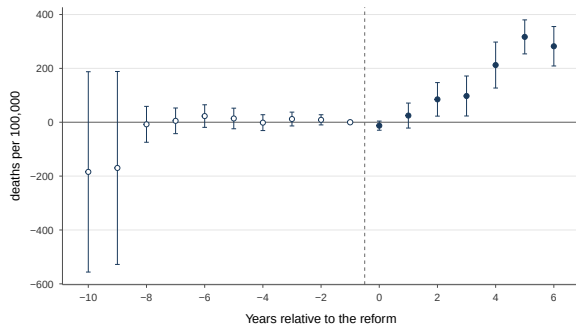
(b) Cerebrovascular disease



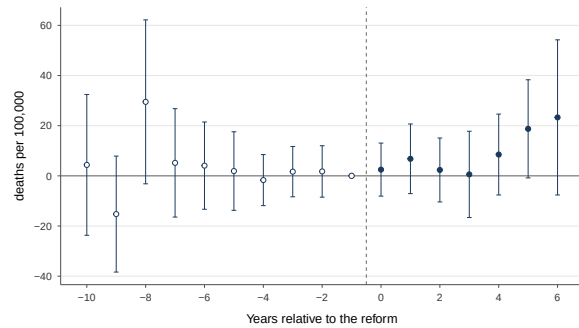
(c) Diabetes



(d) Cancer

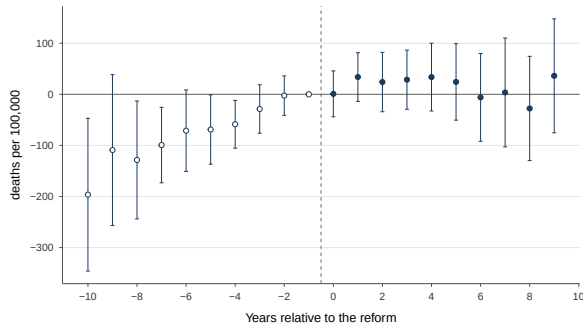


(e) Chronic liver disease

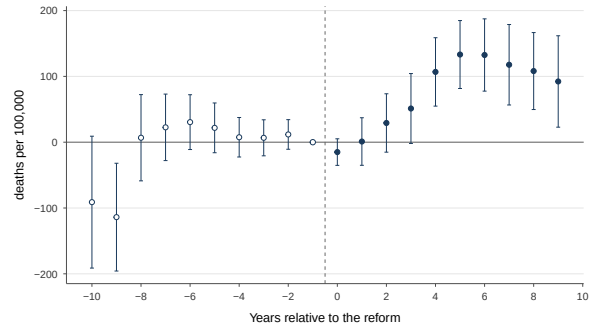


(f) Accidents

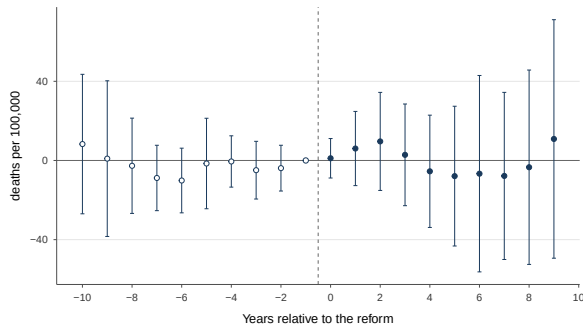
Never-treated comparison, mortality by cause



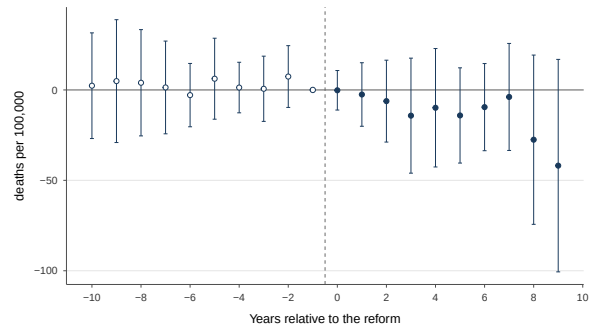
(a) All causes



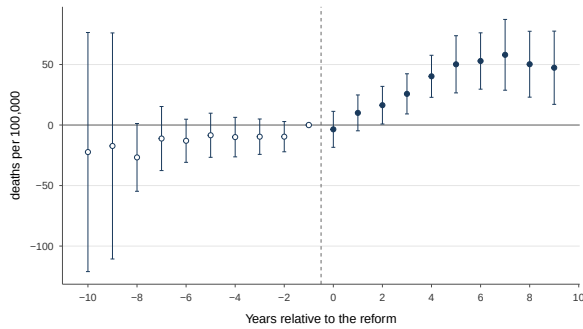
(b) Heart disease



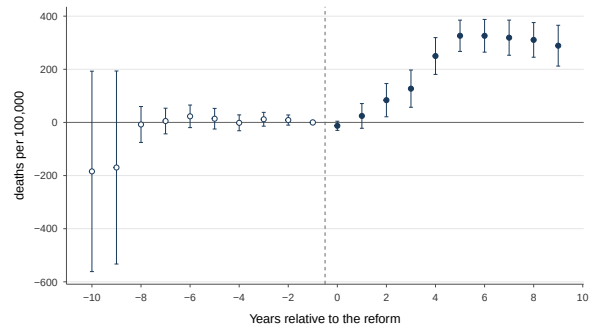
(c) Cerebrovascular disease



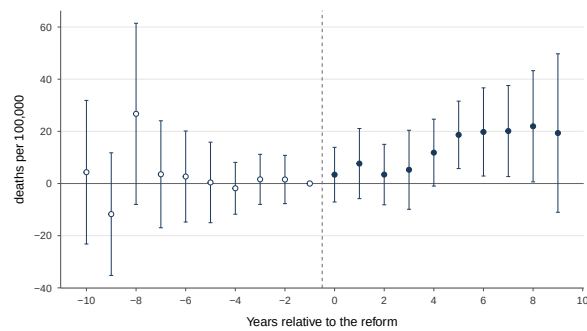
(d) Diabetes



(e) Cancer



(f) Chronic liver disease



(g) Accidents

Never-treated comparison through 2003